EXACT COMPLETION-WEIGHT LAW FOR FORESTS OF A STOPPED COVERING WALK

HUGH ZHENG

Editorial note. Complete original mathematical body and all essential nonadjacent proof components retained. Author owns correctness; no repaired proof, changed hypothesis or independent certification.

Editorial note. Original SIAM class and bibliography style are supplied as local author assets and native class/options retained by the preamble. Source0 omits only pdfTeX-specific pdfoutput switch for XeLaTeX; raw source preserved.

Editorial note. Native disclosure1648–1652 says AI assisted language editing and coding; author takes responsibility for mathematics and formalization correspondence. Original Lean-formalization discussion is retained as author context, without additional certification work.

Editorial note. Cited Aldous-Broder theorem and direct conditional-order/real-aggregation corollaries are context only. General real-matrix nonsingularity assumptions and one-way-support continuity argument remain intact.

Editorial note. Revision r002 also supplies original breakurl.sty and corresponding dtx/ins with upstream LPPL1.2+ copyright/license/provenance for the next actual native-SIAM dependency. Original package/class/mathematics unchanged; no global installation.

Editorial note. XeLaTeX rendering uses the native SIAM class option nohypdvips, which the original class explicitly provides, to disable its dvips-only hyperlink extension. Original class file and mathematical source unchanged; no class patch or replacement.

Editorial note. XeLaTeX presentation passes breakurl its native preserveurlmacro option and removes only breakurl PostScript first-page hook through the LaTeX hook interface. Native hyperref handles URLs; original SIAM class and original breakurl code are unmodified. The class native nohypdvips option disables the dvips-only extension. No mathematical or statement change.

Editorial note. Native siamplain.bst remains the original style; the native receipt build supplies BSTINPUTS pointing to this job local items directory because BibTeX runs in a separate build directory. This is a process-local search path, not a global installation or style replacement.

Allowed standard prerequisites (editorial).

- Finite Markov chains, rooted forests, harmonic extension matrices and adjugates. Native irreducibility/support conditions for walk laws and full nonsingular-base hypothesis for the real-matrix identity retained, along with growth-order definitions, forest/root-block indicators and auxiliary cofactor proofs.. Entire original mathematical body, supplemental algebraic examples, figures and bibliography preserved. The general-chain Aldous-Broder theorem is credited to its existing source and used as context, not counted as a new separate theorem.

1. Introduction. Random spanning forests connect graph structure to statistical inference. In Bayesian clustering, each component of a forest defines a group of observations. A probability distribution over forests then induces a distribution over partitions and quantifies uncertainty in the clustering estimate [16, 7]. In graph signal processing, random forests are used to estimate regularized and interpolated signals by Monte Carlo [20]. Both settings require an understanding of the distribution produced by the sampling procedure. For example, a sample average of a tree-size functional estimates its expectation under the sampler's law, which need not be the forest law specified by the model.

Random walks give a simple way to generate trees. Start at one vertex, run the walk until every vertex has been visited, and retain the edge used to enter each new vertex. These edges form a spanning tree. This is the Aldous–Broder construction [4, 1], one of the classical exact tree samplers alongside Wilson's algorithm [29, 24]; recent methods also use fast-forwarded walks [25]. The tree distribution is known. Our first question is how to obtain that distribution by summing the probabilities of the possible discovery orders directly. Our second question concerns a related forest construction: prescribe a set of roots and stop when all nonroot vertices have been visited. We determine the forest law of this stopped walk and its implications for Monte Carlo estimation.

To describe the problem, let P be the transition matrix of an irreducible Markov chain on $V = \{1, \dots, n\}$, $n \geq 2$, with stationary distribution π. Start the chain at $r \in V$ and run it until every state has been visited. For each $v \neq r$, let $f(v)$ be the state from which v was entered for the first time. The edges $(f(v), v)$ form a spanning arborescence F pointing away from r, called the *first-entrance tree*. Its law is

$$\mathbb{P}(F = F_0) = \frac{\prod_{v \neq r} \pi_{f_{F_0}(v)} P_{f_{F_0}(v)\, v}}{\sum_{F'} \prod_{v \neq r} \pi_{f_{F'}(v)} P_{f_{F'}(v)\, v}}, \tag{1.1}$$

where the sum is over all arborescences F' rooted at r and $f_{F'}$ denotes the parent map of F'. For simple random walk on a connected undirected graph, every edge has the same stationary flow, so (1.1) is uniform over spanning trees. For a general irreducible chain started at a fixed state, (1.1) is the directed Aldous–Broder theorem of Hu, Lyons and Tang [11, Thm. 3.4] and [9, Thm. 3]. The nonreversible result of Fredes and Marckert [9, Thm. 3] assumes that the kernel is positive on the edges of a connected undirected graph. Thus its support assumption excludes one-way transitions, which are allowed here. The general tree law itself is not a new result, and our contribution is the direct evaluation of the sum over discovery orders.

Existing proofs of (1.1) use several approaches. Broder [4] and Aldous [1] use stationary walk constructions; Lyons and Peres [17, Cor. 4.9] give a weighted reversible version. For general chains, Hu, Lyons and Tang [11] group finite paths by directed transition counts and use an Eulerian-path correspondence. Fredes and Marckert [9] develop a combinatorial encoding based on cycles and paths. A direct calculation instead conditions on the order in which the vertices of a fixed tree are discovered. The difficulty of summing over these orders is discussed by Fredes and Marckert [9, §1 and Remark 19]: each factor depends on the set visited at that stage. Two orders of the same tree can therefore have different probabilities, and evaluating one order does not evaluate their sum.

We now make this summation problem explicit. Let $\tau(X)$ be the order in which the states other than r are first entered. For an ordering $\tau = (\tau_1, \dots, \tau_{n-1})$ compatible with a fixed tree F_0, let B_ℓ be the set visited before the $(\ell + 1)$-st first entrance.

Thus $B_0 = \{r\}$ and $B_\ell = B_{\ell-1} \cup \{\tau_\ell\}$. Between consecutive first entrances, the walk stays inside the visited set. Summing over the lengths of these excursions gives (Proposition 3.2)

$$\mathbb{P}\big(F = F_0,\ \tau(X) = \tau\big) = \kappa \cdot w(F_0) \cdot \prod_{\ell=1}^{n-2} h_\ell(f, \tau), \tag{1.2}$$

where f is the parent map of F_0, $w(F_0) := \prod_{v \neq r} \pi_{f(v)} P_{f(v)v}$ is its weight, and κ depends only on the chain. Each factor $h_\ell(f, \tau)$ has a probabilistic interpretation. If $f(\tau_{\ell+1}) \in B_{\ell-1}$, it is the probability that the reversed chain started at that parent first leaves $B_{\ell-1}$ at τ_ℓ. If the parent is τ_ℓ itself, the factor is one. The tree weight is common to all compatible orders. Hence the tree law follows once we establish

$$\sum_{\tau:\tau \text{ compatible with } F_0} \prod_{\ell=1}^{n-2} h_\ell(f, \tau) = 1.$$

for each fixed tree. This normalization does not follow by treating the tree and its discovery order as independent, since they are generally dependent. It requires evaluating terms that involve different sequences of visited sets.

Our approach is to express each factor $h_\ell(f, \tau)$ through a harmonic extension of $L := I - \Pi^{-1} P^\top \Pi = I - P^*$, where $\Pi := \operatorname{diag}(\pi)$ and P^* is the transition matrix of the reversed chain. This representation turns the probability calculation into a matrix identity. The identity holds for any real matrix with zero row sums, provided that the principal submatrices needed for the harmonic extensions are nonsingular. To formulate it, we allow the fixed graph to be a rooted forest. Its *order mass* at a root is the sum of the order weights over growth orders whose first vertex is a child of that root. Theorem 4.5 identifies this mass with a principal cofactor of a matrix indexed by the roots. Consequently, the sum depends only on the partition into root blocks, although the individual terms depend on the edges inside each block.

With one root, the cofactor is the determinant of the empty matrix, which equals one. This gives the normalization in (1.2) and a direct proof of the tree law. With several roots, the identity gives a closed-form evaluation of the completion weight in the stopped-forest law. Although the law itself is a marginal of (1.1), the calculation makes its extra factor, which we call completion weight, explicit. Forests with large completion weights are favored, and some forests of positive target weight may never be sampled. Inverse-completion weighting recovers the target law conditional on the reachable forests. It recovers the full target law only when no target mass is lost. We illustrate these distinctions on small chains whose laws can be enumerated completely.

The identity also connects to classical matrix formulas. Matrix-tree theorems expand cofactors into tree and forest sums [27, 5], while harmonic tree formulas express exit probabilities as ratios of forest sums [21]. Determinant ratios also occur in the study of loop-erased walks [18, 8]. Our sum is indexed by the growth orders of a fixed forest, and each term uses harmonic extensions on a sequence of growing sets. We evaluate it by moving one vertex into the root set and comparing the corresponding cofactors. This step explains why the several-root identity is needed even to prove the single-root normalization.

The article makes four contributions.

1. We establish the order-mass identity under a condition involving only the proper bases of the given forest's growth orders (Theorem 4.5). The proof uses finite linear algebra and extends to any field. A special case gives a weighted count of the linear extensions of a forest (Corollary 4.9).

2. We use the identity to give a direct proof of the extended Aldous–Broder theorem for every finite irreducible chain from a fixed start, and derive the conditional distribution of its discovery order (Theorems 6.1 and 6.2).
3. We evaluate the completion weight in the law of the forest produced by the stopped covering walk (Theorem 6.4). The evaluation gives an explicit comparison with the weight-proportional law and identifies the support condition for importance correction.
4. We formalize the identity, the Aldous–Broder theorem and the stopped-walk law in Lean 4 [6] on top of mathlib [26]. To our knowledge, this is the first machine-checked proof of the Aldous–Broder theorem.

The remainder of the article follows the calculation. Section 2 introduces harmonic extensions and forest growth orders. Section 3 computes the probability of a fixed forest and discovery order. Section 4 states the order-mass identity, and Section 5 proves it by transferring one vertex into the root set. Section 6 derives the tree, order, and stopped-forest laws. The appendices provide worked examples, algebraic extensions, the formalization correspondence, and numerical illustrations.

2. Preliminaries. The calculation uses two objects: harmonic extensions on growing sets of vertices, and orders compatible with a fixed forest. We first give the notation and cofactor identities needed to define them.

2.1. Notation. Throughout, V is a set of $n \geq 2$ vertices, and $U^c := V \setminus U$ for $U \subseteq V$. We use u, v, w for vertices and r for a root. For a matrix A, we write $A_{\cdot v}$ for its column v, $A_{u\cdot}$ for its row u, $A[U]$ for the principal submatrix on the rows and columns in U, $A[R, C]$ for the submatrix on the rows R and the columns C, and $\operatorname{adj}(A)$ for the adjugate of a square matrix, the transpose of its cofactor matrix. Matrices and vectors are indexed by sets of vertices. For example, if $U = \{1, 2\}$ and $V = \{1, 2, 3\}$, a matrix $A \in \mathbb{R}^{U \times U^c}$ has one column, indexed by 3; thus $A_{\cdot 3}$ denotes that column.

The *principal cofactor* of $A \in \mathbb{R}^{U \times U}$ at $u \in U$ is $\det(A[U \setminus \{u\}])$. We use e_v for the standard basis vectors, $\mathbf{1}$ for the all-ones vector and $\mathbf{0}$ for the zero vector. We write $x|_U$ for the restriction of a vector x to U, and we read a vector given on a subset of V as a vector on V with the missing entries 0 when the context requires it. For a statement E, $[E]$ is 1 if E holds and 0 otherwise.

2.2. Preliminary definitions and results. We use three classical facts about determinants [10, §§0.8.2–0.8.5]: for $A \in \mathbb{R}^{U \times U}$ and $a, c \in \mathbb{R}^U$,

$$\begin{gathered} A \operatorname{adj}(A) = \operatorname{adj}(A) A = \det(A) I, \qquad \det(I + ac^\top) = 1 + c^\top a, \\ \det(A \text{ with column } v \text{ replaced by } a) = \textstyle\sum_u a_u \operatorname{adj}(A)_{vu}. \end{gathered} \tag{2.1}$$

We will also need two facts about the cofactors of a matrix with zero row sums.

LEMMA 2.1 (Cofactors of a matrix with zero row sums). *Let S be a nonempty finite set, let $A \in \mathbb{R}^{S \times S}$ satisfy $A\mathbf{1} = \mathbf{0}$, and let $d \in \mathbb{R}^S$ be its vector of principal cofactors, $d_u := \det(A[S \setminus \{u\}])$. Then*

(a) $\operatorname{adj}(A) = \mathbf{1} d^\top$ *and* $d^\top A = \mathbf{0}^\top$;

(b) $\det(A + ac^\top) = (c^\top \mathbf{1})(a^\top d)$ *for all* $a, c \in \mathbb{R}^S$.

Part (a) identifies the cofactor row as a left null vector. Part (b) evaluates the determinant after a rank-one change through that row. Together, these facts allow us to compare cofactors before and after one vertex is added to a base in section 5. The proof is in section B.

2.3. Harmonic extensions. Suppose that values are given on the complement of U. A harmonic extension fills in the values on U so that the corresponding rows of L annihilate the resulting vector.

DEFINITION 2.2 (Harmonic extension). *Let $L \in \mathbb{R}^{V\times V}$, let $U \subsetneq V$ with $\det L[U] \neq 0$, and let $m \in \mathbb{R}^{U^c}$. We call the matrix $\tilde{\Gamma}^U \in \mathbb{R}^{V\times U^c}$ with the rows*

$$\tilde{\Gamma}^U[U, U^c] := -L[U]^{-1}L[U, U^c], \qquad \tilde{\Gamma}^U[U^c, U^c] := I_{|U^c|}$$

the harmonic extension map*, and $\gamma^U(m) := \tilde{\Gamma}^U m \in \mathbb{R}^V$ the* harmonic extension *of m into U. In particular, $\tilde{\Gamma}^\emptyset = I$ and $\gamma^\emptyset(m) = m$.*

The harmonic extension is characterized by two properties.

LEMMA 2.3. *Let $L \in \mathbb{R}^{V\times V}$, let $U \subsetneq V$ with $\det L[U] \neq 0$, and let $m \in \mathbb{R}^{U^c}$. For $x \in \mathbb{R}^V$,*

$$x|_{U^c} = m \quad \text{and} \quad (Lx)|_U = \mathbf{0} \qquad \Longleftrightarrow \qquad x = \gamma^U(m).$$

Proof. We have $(Lx)|_U = L[U]\, x|_U + L[U, U^c]\, x|_{U^c}$. Hence, given $x|_{U^c} = m$, the condition $(Lx)|_U = \mathbf{0}$ holds if and only if $x|_U = -L[U]^{-1}L[U, U^c]m$, that is, if and only if $x = \tilde{\Gamma}^U m$. This completes the proof. ∎

Remark 2.4. By Lemma 2.3, $\gamma^U(m)$ is the unique vector in $\mathbb{R}^V$ that equals m off U and is annihilated by the rows U of L.

The map $m \mapsto \gamma^U(m) = \tilde{\Gamma}^U m$ is linear. Each column has a simple interpretation: for $v \in U^c$, the vector $\gamma^U(e_v)$ equals 1 at v and 0 at the other vertices outside U. Its restriction to U solves the Dirichlet problem with these boundary values. If $L\mathbf{1} = \mathbf{0}$, then $\mathbf{1}$ is annihilated by every row of L and equals $\mathbf{1}$ off U; hence, by uniqueness,

$$\gamma^U(\mathbf{1}) = \mathbf{1}, \qquad \text{i.e.} \qquad \sum_{v\in U^c} \gamma^U(e_v) = \mathbf{1}. \tag{2.2}$$

For ease of notation, we write

$$\Gamma^U := \tilde{\Gamma}^U[U, U^c] = -L[U]^{-1}L[U, U^c] \in \mathbb{R}^{U\times U^c} \tag{2.3}$$

for the rows of $\tilde{\Gamma}^U$ indexed by U. If $L\mathbf{1} = \mathbf{0}$, then $\Gamma^U\mathbf{1} = \mathbf{1}$ by (2.2).

2.4. Forests, blocks, and growth orders. To specify a forest, we record the parent of each nonroot vertex. An order is compatible with this forest when each parent is available before its child is added. We call these compatible orders growth orders. The parent map also partitions the nonroot vertices according to the root of their tree.

DEFINITION 2.5 (Forests, growth orders, root indicator). *Let $B \subsetneq V$ be nonempty and write $k := |B^c|$.*

(a) A map $f : B^c \to V$ is a B-parent map *if, for every $v \in B^c$, some iterate $f^{(q)}(v)$ lies in B, where $f^{(0)}(v) := v$ and $f^{(q+1)}(v) := f(f^{(q)}(v))$ as long as $f^{(q)}(v) \in B^c$. The vertex $f(v)$ is the* parent *of v, the elements of $f^{-1}(u)$ are the* children *of u, and the* block *of $u \in V$ is $D_u := \{v \in B^c : f^{(q)}(v) = u \text{ for some } q \geq 0\}$.*

(b) A growth order *of a B-parent map f is an ordering $\tau = (\tau_1, \dots, \tau_k)$ of B^c with $f(\tau_\ell) \in B_{\ell-1}$ for every ℓ, where*

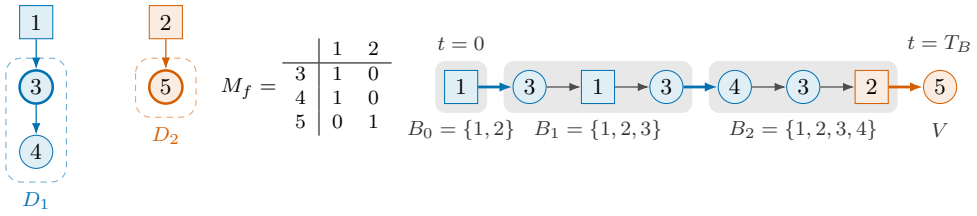
$$B_\ell := B \cup \{\tau_1, \dots, \tau_\ell\} \qquad (0 \leq \ell \leq k) \tag{2.4}$$

is the base *after ℓ steps. We write $\mathcal{S}_B(f)$ for the set of growth orders of f.*

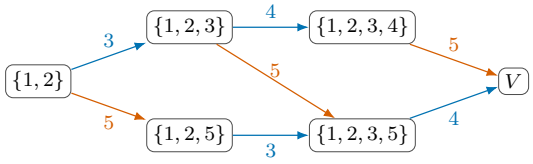
(c) *The* root indicator *of a B-parent map f is the matrix $M_f \in \{0,1\}^{B^c \times B}$ with entries $(M_f)_{vr} = 1$ if $v \in D_r$ and 0 otherwise.*

The edge set $\{(f(v), v) : v \in B^c\}$ of a B-parent map f is a spanning forest of V rooted at B, with each tree pointing away from its root, and f and the forest determine each other. We write $\mathcal{F}_B$ for the set of these forests, and $\mathcal{F}_r := \mathcal{F}_{\{r\}}$ for the spanning arborescences rooted at r.

Figure 1 illustrates Definition 2.5 using a simple example. There $V = \{1, \dots, 5\}$, the roots are $B = \{1, 2\}$ (squares), and the B-parent map is $f(3) = 1$, $f(4) = 3$, $f(5) = 2$; the tree of the root 1 is drawn in blue and that of the root 2 in vermilion. In panel (a), the children of the roots are 3 and 5 (thick circles), and the dashed regions are the root blocks $D_1 = \{3, 4\}$ and $D_2 = \{5\}$. The ordering $(3, 5, 4)$ is a growth order; $(4, 3, 5)$ is not, since $f(4) = 3 \notin B$. Panel (b) shows a path of a stopped covering walk on V, which is a walk $X_0, \dots, X_7$ from the seed 1. Thick steps are first entrances, and the gray blocks are the excursions between them; the ℓ-th excursion stays inside B_ℓ, where $B_0 := B = \{1, 2\}$, $B_1 = \{1, 2, 3\}$ and $B_2 = \{1, 2, 3, 4\}$. The first-entrance forest is f, and the discovery order is $(3, 4, 5)$. In panel (c) we illustrate the growth orders as paths. Each edge between two bases adds the vertex on its label. The three paths are the growth orders $(3, 4, 5)$, $(3, 5, 4)$ and $(5, 3, 4)$. Intuitively, the growth orders are the orders in which the forest can be grown one vertex at a time, with each new vertex a child of a present one; equivalently, they are the linear extensions of the tree order of f.



(a) Forest and root indicator (b) Stopped covering walk



(c) Growth orders as paths

FIG. 1. *An illustration of Definition 2.5 using a simple example.*

We now record the only structural fact that the proofs need.

LEMMA 2.6 (Blocks). *Let f be a B-parent map. The blocks D_r, $r \in B$, partition B^c, and for every $u \in V$ the blocks of the children of u are pairwise disjoint with union $D_u \setminus \{u\}$.*

As a consequence of Lemma 2.6, $M_f \mathbf{1} = \mathbf{1}$; that is, the root indicator records, for each non-root, the root of its tree, and nothing else about the forest. We defer the proof of Lemma 2.6 to section B.

3. The covering walk and the sum over discovery orders. We now connect the linear algebra to the random walk. The main calculation separates the probability of a fixed forest and discovery order into three parts: a normalizing constant, the forest weight, and an order weight. The last part is a product of harmonic extension entries for the reversed chain. This decomposition identifies the sum that the order-mass identity must evaluate.

3.1. The walk and its first-entrance forest. Let P be an irreducible stochastic matrix on V, and let π be its stationary distribution, with $\pi_v > 0$, $\mathbf{1}^\top \pi = 1$ and $\pi^\top P = \pi^\top$ [14, Ch. 1]. We use $(X_t)_{t\geq 0}$ to denote the Markov chain with transition matrix P. Throughout this section and section 6, we consider

$$L := I - \Pi^{-1} P^\top \Pi, \qquad \Pi := \operatorname{diag}(\pi), \tag{3.1}$$

so that $L_{uv} = [u = v] - P^*_{uv}$ with $P^*_{uv} := \pi_v P_{vu}/\pi_u$, where P^* is the transition matrix of the reversed chain. The rows of L sum to zero, since π is stationary.

For this matrix, harmonic extensions describe exit probabilities of the reversed chain. Fix $U \subsetneq V$ and $v \in U^c$, and let x_u be the probability that the reversed chain started at $u \in U$ first leaves U at v. Set $x_v = 1$ and $x_w = 0$ for the other vertices outside U. First-step conditioning gives $(Lx)|_U = \mathbf{0}$. The chain exits U almost surely by finite irreducibility, and $L[U]$ is nonsingular by Lemma 3.3. Hence Lemma 2.3 identifies x with $\gamma^U(e_v)$. In particular, Γ^U is the exit-distribution matrix from U.

We now fix a nonempty set of roots $B \subsetneq V$ and a *seed* $r \in B$. We start the chain at $X_0 = r$, and let $T_v := \min\{t \geq 0 : X_t = v\}$ and $T_B := \max_{v\in B^c} T_v$, the first time every non-root has been visited; all of these are almost surely finite [14, Lem. 1.13]. The *first-entrance forest* is

$$F_B := \{(X_{T_v - 1}, v) : v \in B^c\},$$

that is, the set of first-entrance edges of the non-roots, while the roots may be visited freely. Since following parents strictly decreases the first-entrance time, we have $F_B \in \mathcal{F}_B$, and its parent map sends v to X_{T_v-1}. We write $\tau(X)$ for the order in which the walk first enters the vertices of B^c, and call it the *discovery order*. Since the parent of a vertex is entered before the vertex, $\tau(X)$ is a growth order of the parent map of F_B. Algorithm 3.1 gives the construction as a procedure. For $B = \{r\}$, it is the Aldous–Broder algorithm, and F_B is the first-entrance tree F of (1.1); for larger B, we refer to it as the *stopped covering walk*. The root set is fixed in advance and serves only the bookkeeping of the forest: the walk starts at the seed alone, it may or may not visit the other roots, and it stops when every vertex of B^c has been visited. Thus stopping early refers to covering the prescribed set B^c, not to stopping at a fixed time or after a fixed number of discoveries. For clarity, Figure 1(b) illustrates the walk, its visited sets and the resulting forest.

Algorithm 3.1 Stopped covering walk and its first-entrance forest.

Input: an irreducible stochastic matrix P on V; a nonempty root set $B \subsetneq V$; a seed $r \in B$.

Output: a spanning forest $F_B \in \mathcal{F}_B$, each edge $(f(v), v)$ pointing away from the root of its tree.

$X \leftarrow r$; $U \leftarrow B$; $F_B \leftarrow \emptyset$
while $U \neq V$ **do**
  draw Y from the distribution P_X.
  if $Y \notin U$ **then**
    $F_B \leftarrow F_B \cup \{(X, Y)\}$; $U \leftarrow U \cup \{Y\}$
  $X \leftarrow Y$
return F_B

The forest and discovery order contribute different factors to the probability of the walk. We define their weights separately.

Definition 3.1 (Forest weight and order weight). *Let f be a B-parent map, and let $F_0 \in \mathcal{F}_B$ be its forest.*

(a) The forest weight *of F_0 is*

$$w(F_0) := \prod_{v \in B^c} \pi_{f(v)} P_{f(v)v} = \Big(\prod_{v \in B^c} \pi_v \Big) \prod_{v \in B^c} P^*_{vf(v)}. \tag{3.2}$$

(b) Let $L \in \mathbb{R}^{V \times V}$, and let $\tau \in \mathcal{S}_B(f)$ satisfy $\det L[B_\ell] \neq 0$ for $0 \leq \ell \leq k-1$. The order weight *of τ is*

$$\omega_f(\tau) := \prod_{\ell=1}^{k-1} \tilde{\Gamma}^{B_{\ell-1}}_{f(\tau_{\ell+1}),\tau_\ell}, \tag{3.3}$$

where an empty product is 1. It depends on L and f; we suppress L from the notation.

The forest weight is proportional to the product of reversed transitions from each vertex to its parent. The order weight records how the next parent is reached as the visited set grows. When L is the matrix (3.1), its factors are exactly the $h_\ell(f, \tau)$ of (1.2). To see this, note that the parent $f(\tau_{\ell+1})$ of the next vertex lies in B_ℓ. If it lies in $B_{\ell-1}$, the factor at ℓ is the probability that the reversed chain started there first leaves $B_{\ell-1}$ at the vertex τ_ℓ just added. Otherwise, the parent is τ_ℓ, and the factor is 1, since $\gamma^U(e_v)$ is 1 at v. Hence only the steps at which the next vertex hangs from an earlier vertex contribute. For example, the growth order $(3, 5, 4)$ in Figure 1 has the two factors $\Gamma^{B_0}_{2,3}$ and $\Gamma^{B_1}_{3,5}$, while $(3, 4, 5)$ has the single factor $\Gamma^{\{1,2,3\}}_{2,4}$ and $(5, 3, 4)$ the single factor $\Gamma^{\{1,2\}}_{1,5}$.

Let

$$\lambda_v := \det\big((I - P)[V \setminus \{v\}]\big) \quad (v \in V), \qquad \lambda := (\lambda_v)_{v \in V}. \tag{3.4}$$

We use the principal cofactors λ to express the common normalizing constant in the following proposition.

3.2. The order decomposition.

PROPOSITION 3.2 (Order decomposition). *Let P be irreducible and stochastic with stationary distribution π, let L and λ be as in (3.1) and (3.4), let $B \subsetneq V$ be nonempty with seed $r \in B$, and let F_B and $\tau(X)$ be the first-entrance forest and discovery order of the stopped covering walk. For every $F_0 \in \mathcal{F}_B$ with parent map f and every $\tau \in \mathcal{S}_B(f)$,*

$$\mathbb{P}\big(F_B = F_0,\ \tau(X) = \tau\big) = \frac{\det L[B]\,\big(L[B]^{-1}\big)_{f(\tau_1),\,r}\, w(F_0)\,\omega_f(\tau)}{\pi_r\big(\prod_{v\in B^c}\pi_v\big)\mathbf{1}^\top\lambda}, \tag{3.5}$$

and $\mathbb{P}(F_B = F_0,\ \tau(X) = \tau) = 0$ for every ordering τ of B^c that is not a growth order of f. In particular, for $B = \{r\}$,

$$\mathbb{P}\big(F = F_0,\ \tau(X) = \tau\big) = \frac{w(F_0)\cdot\omega_f(\tau)}{(\mathbf{1}^\top\lambda)\prod_{v\in V}\pi_v}. \tag{3.6}$$

The proof has two ingredients. A Green function sums the excursions inside each visited set. Reversal and a normalized inverse column then express that Green function through a harmonic extension. The following lemmas give these identities and their normalization.

LEMMA 3.3 (Green function, reversal, normalized columns). *Let P be irreducible and stochastic with stationary distribution π, and let L be the matrix (3.1). Then $L\mathbf{1} = \mathbf{0}$ and $\det L[U] = \det\big((I-P)[U]\big) > 0$ for every nonempty $U \subsetneq V$. Moreover, for $u, v \in U$,*

$$\Big(\big((I-P)[U]\big)^{-1}\Big)_{uv} = \sum_{t\ge 0}\big(P[U]^t\big)_{uv} = \frac{\pi_v}{\pi_u}\big(L[U]^{-1}\big)_{vu}, \tag{3.7}$$

and, if $|U| \ge 2$, the column v of $L[U]^{-1}$ normalized by its diagonal entry is a harmonic extension:

$$\big(L[U]^{-1}\big)_{vv} = \frac{\det L[U\setminus\{v\}]}{\det L[U]}, \qquad \frac{L[U]^{-1}e_v}{(L[U]^{-1})_{vv}} = \gamma^{U\setminus\{v\}}(e_v)\big|_U. \tag{3.8}$$

The Neumann series gives the Green function, and Cramer's rule gives the normalized column. We provide the proof in section B.

LEMMA 3.4 (Cofactor form of the stationary distribution). *Let P be irreducible and stochastic with stationary distribution π. Then $\lambda_v > 0$ for all v, and $\pi = \lambda/(\mathbf{1}^\top\lambda)$.*

Proof of Lemma 3.4. Since $I - P$ has zero row sums, Lemma 2.1(a) gives $\lambda^\top(I - P) = \mathbf{0}^\top$, and each λ_v is positive by Lemma 3.3. Hence $\lambda/(\mathbf{1}^\top\lambda)$ is a stationary distribution, and the uniqueness of the stationary distribution of an irreducible chain [14, Ch. 1] completes the proof. ∎

Proof of Proposition 3.2. An ordering that is not a growth order has probability zero, since the walk visits a parent before it first enters its child. We therefore fix $\tau \in \mathcal{S}_B(f)$, put $\tau_0 := r$, and let B_ℓ be the bases (2.4); $B_{\ell-1}$ is the set of vertices that may be visited before the ℓ-th first entrance into B^c. On the event $\{F_B = F_0, \tau(X) = \tau\}$, for $\ell = 1, \dots, k$, the walk moves inside $B_{\ell-1}$ from $\tau_{\ell-1}$ to $f(\tau_\ell)$ and then steps to τ_ℓ. By the Markov property, summing over the lengths of these confined excursions and using (3.7), we have

$$\mathbb{P}\big(F_B = F_0,\ \tau(X) = \tau\big) = \prod_{v\in B^c} P_{f(v)v}\cdot\prod_{\ell=1}^{k}\Big(\big((I-P)[B_{\ell-1}]\big)^{-1}\Big)_{\tau_{\ell-1},\,f(\tau_\ell)}.$$

By the reversal in (3.7), the factor ℓ equals $(\pi_{f(\tau_\ell)}/\pi_{\tau_{\ell-1}})(L[B_{\ell-1}]^{-1})_{f(\tau_\ell),\tau_{\ell-1}}$. Here the numerators complete $\prod_{v\in B^c} P_{f(v)v}$ to the forest weight $w(F_0)$, and the denominators multiply to $\prod_{\ell=1}^{k} \pi_{\tau_{\ell-1}} = \pi_r\big(\prod_{v\in B^c}\pi_v\big)/\pi_{\tau_k}$. For $2 \le \ell \le k$, applying (3.8) with $U := B_{\ell-1}$ and $v := \tau_{\ell-1}$, so that $U \setminus \{v\} = B_{\ell-2}$, we have

$$\big(L[B_{\ell-1}]^{-1}\big)_{f(\tau_\ell),\,\tau_{\ell-1}} = \big(\gamma^{B_{\ell-2}}(e_{\tau_{\ell-1}})\big)_{f(\tau_\ell)} \cdot \frac{\det L[B_{\ell-2}]}{\det L[B_{\ell-1}]}.$$

The product of these ratios simplifies to $\det L[B]/\det L[V\setminus\{\tau_k\}]$, where

$$\det L[V\setminus\{\tau_k\}] \overset{(a)}{=} \lambda_{\tau_k} \overset{(b)}{=} \pi_{\tau_k}\mathbf{1}^\top\lambda,$$

and (a) and (b) follow from Lemma 3.3 and Lemma 3.4, respectively. The factor $\ell = 1$ is the entry $(L[B]^{-1})_{f(\tau_1),r}$. Finally, the factor π_{τ_k} cancels, and the harmonic factors, reindexed by $\ell \mapsto \ell - 1$, form $\omega_f(\tau)$; this gives (3.5). For $B = \{r\}$, we have $\det L[B] = L_{rr}$ and $(L[B]^{-1})_{rr} = 1/L_{rr}$, which gives (3.6). This completes the proof. □

Remark 3.5 (The Markov chain tree theorem). Lemma 3.4 is the linear-algebra half of the Markov chain tree theorem [28, Lem. 7.1], [13, 2]. Only this half is needed in what follows; the expansion of λ_v into arborescence sums plays no role.

The one-step factor of this decomposition, the law of the next first-entrance edge given the visited set and the current state, is also the step of the fast-forwarded sampler of Tam, Dunson and Duan [25]. What the order decomposition adds is the product over a whole growth order, written through the order weight. Summing (3.6) over $\tau \in \mathcal{S}_{\{r\}}(f)$, we see that (1.1) follows as soon as

$$\sum_{\tau\in\mathcal{S}_{\{r\}}(f)} \omega_f(\tau) = 1 \tag{3.9}$$

for every arborescence of positive weight. In the next section, we evaluate the left side of (3.9) for every matrix with zero row sums and nonvanishing minors, and every number of roots.

4. The order-mass identity. The order decomposition leaves a deterministic problem: sum the order weights of a fixed forest. We study this problem for an arbitrary real matrix $L \in \mathbb{R}^{V\times V}$ with zero row sums. Fix a nonempty root set $B \subsetneq V$, write $k := |B^c|$, and let f be a B-parent map. The only nonsingularity conditions concern the bases encountered along the growth orders of this forest.

Assumption 4.1 (Nonsingular bases). *$L\mathbf{1} = \mathbf{0}$, and $\det L[B_\ell] \neq 0$ for every growth order $\tau \in \mathcal{S}_B(f)$ and every $0 \le \ell \le k-1$.*

Remark 4.2. Assumption 4.1 ensures that each Dirichlet problem along the growth of f has a unique solution. It concerns only the bases B_ℓ of (2.4), which are determined by f. The assumption is satisfied for every B and f when L is the matrix (3.1) of an irreducible chain, by Lemma 3.3, and when L is the Laplacian of a connected undirected graph with positive weights, whose proper principal submatrices are positive definite. In general, by the all-minors matrix-tree theorem [5], $\det L[U]$ is a signed sum over the forests rooted at U^c; hence the assumption asks that these forest sums do not vanish.

Definition 4.3 (Order mass). *Under Assumption 4.1, the* order mass *of f at the root $r \in B$ is*

$$g_r(B,f) := \sum_{\tau\in\mathcal{S}_B(f):\ f(\tau_1)=r} \omega_f(\tau), \tag{4.1}$$

the total order weight (3.3) *of the growth orders whose first vertex is a child of* r*. We write* $g(B,f) := (g_r(B,f))_{r\in B}$*.*

Remark 4.4. The name comes from the chain case, in which the order weights are products of probabilities. With one root, the order mass equals one, and the order weights form the conditional law of the discovery order given the tree (Theorem 6.2). With several roots, however, the order masses are not necessarily normalized; see Theorem 4.6.

THEOREM 4.5 (Order-mass identity). *Let* $L \in \mathbb{R}^{V\times V}$*,* B *and* f *satisfy Assumption 4.1. Then, for every* $r \in B$*,*

$$g_r(B,f) = \det\big((I-\Gamma^B M_f)[B\setminus\{r\}]\big), \tag{4.2}$$

with the determinant of the empty matrix being 1*. In particular, if* $B=\{r\}$*, then the order weights of the growth orders of every arborescence rooted at* r *sum to* 1*.*

Equation (4.2) separates the roles of the forest's edges and its partition into root blocks. On the left, each growth order uses a sequence of bases and depends on the edges inside the trees. On the right, the forest enters only through its root indicator. Thus two forests with the same root blocks have the same order masses. The dependence on the internal tree structure disappears after summing over orders.

Example 4.6 (A two-root and a one-root calculation). Let $L = I - \frac{1}{5}W^\top$ be the matrix of the doubly stochastic chain $P = W/5$ of Example A.1, and take $B=\{1,2\}$ with the forest $f_3 : 3\mapsto 1,\ 4\mapsto 2$. Its two growth orders are $(3,4)$, which enters at the root 1 with the single factor $\Gamma^B_{f_3(4),3} = \Gamma^B_{2,3} = \frac{3}{5}$, and $(4,3)$, which enters at the root 2 with the factor $\Gamma^B_{1,4} = \frac{3}{5}$. Hence $g(B,f_3) = (\frac{3}{5},\frac{3}{5})$. On the right side, $\Gamma^B_{\cdot 3} = (\frac{2}{5},\frac{3}{5})^\top$ and $\Gamma^B_{\cdot 4} = (\frac{3}{5},\frac{2}{5})^\top$, so that

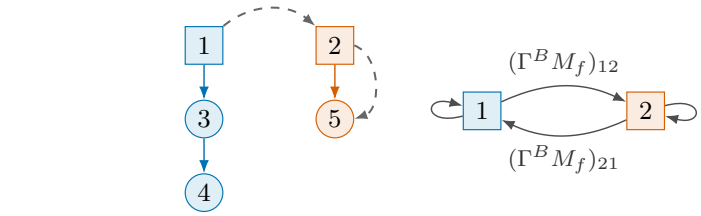
$$I-\Gamma^B M_{f_3} = \begin{pmatrix} \frac{3}{5} & -\frac{3}{5} \\ -\frac{3}{5} & \frac{3}{5} \end{pmatrix},$$

whose principal cofactors are $(\frac{3}{5},\frac{3}{5})$, as Theorem 4.5 asserts. Each of these orders has one factor. With the single root 1 and the star $f_\star : 2,3,4\mapsto 1$, on the other hand, every ordering of $\{2,3,4\}$ is a growth order, the bases change at every step, and the six order weights are

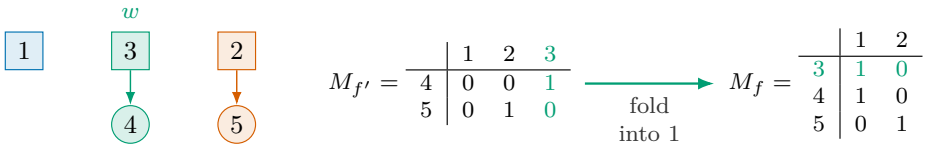
$$\tfrac{1}{10},\ \tfrac{3}{20},\ \tfrac{1}{10},\ \tfrac{3}{20},\ \tfrac{1}{4},\ \tfrac{1}{4} \qquad \text{for } (2,3,4),(2,4,3),(3,2,4),(3,4,2),(4,2,3),(4,3,2),$$

which sum to 1. Section A gives the remaining forests of this chain and the stopped-walk law.

Remark 4.7 (The chain on the roots). The right side of (4.2) is a cofactor of the $B\times B$ matrix $I-\Gamma^B M_f$, whose column u is $e_u - (\Gamma^B M_f)_{\cdot u} = e_u - \sum_{v\in D_u}\Gamma^B_{\cdot v}$. Since Γ^B and M_f have unit row sums, so does $\Gamma^B M_f$; hence $I-\Gamma^B M_f$ has zero row sums, and by Lemma 2.1(a) its principal cofactors form the common row of its adjugate. For a Markov chain, $\Gamma^B M_f$ is stochastic: its entry (u,u') is the probability that the reversed chain started at the root u leaves B into the tree of u'. By Theorem 4.5 and Lemma 2.1(a), the order masses then form an invariant measure of this chain on the roots (Figure 2(a)). This measure need not be normalized, and it vanishes when the chain on the roots has more than one closed class, since $I-\Gamma^B M_f$ then has rank at most $|B|-2$.



(a) Chain on the roots



(b) Moving w into the roots

(c) Folding into r

FIG. 2. *The roots of the forest of Figure 1, with the moved vertex w and its block in green. (a) Chain on the roots (Theorem 4.7). Dashed, a path of the reversed chain from the root 1 that first leaves B into the tree of 2; on the right, the chain on the roots with transition matrix $\Gamma^B M_f$. (b) Moving w into the roots, the induction step of section 5 for $r = 1$ and $w = 3$; the new forest f' is rooted at $B' = \{1, 2, 3\}$. (c) Folding into r (Lemma 5.1): adding the column w of $M_{f'}$ to the column r and inserting the row $e_r^\top$ for w recovers M_f.*

Remark 4.8 (Forest sums). Both sides of (4.2) are ratios of forest sums. By Cramer's rule and the all-minors matrix-tree theorem [5], each entry Γ^U_{uv}, $u \in U$, $v \in U^c$, is the signed sum over the forests rooted at U^c in which u lies in the tree of v, divided by the signed sum over all forests rooted at U^c, each forest counted by the product of the off-diagonal entries of $-L$ along its arcs; in the chain instance, the signs are positive, and this is the harmonic tree formula [21, (1.9)], [23]. On the right, the matrix-tree theorem [27, 5] gives

$$\det\big((I - \Gamma^B M_f)[B \setminus \{r\}]\big) = \sum_T \prod_{u \in B \setminus \{r\}} \big(\Gamma^B M_f\big)_{u,\, T(u)}, \tag{4.3}$$

the sum over the arborescences T on B rooted at r, with $T(u)$ the parent of u. Thus the growth orders entering at r weigh as much as these arborescences. However, no weight-preserving correspondence between the two index sets is known; the left side has $|\{\tau \in \mathcal{S}_B(f) : f(\tau_1) = r\}|$ terms, whereas the right side has at most $|B|^{|B|-2}$.

We now give a special case in which every factor is explicit and the matrix disappears from the statement. Recall that successive sampling without replacement from B^c with positive weights p draws the ordering τ with probability $\prod_{\ell=1}^k p_{\tau_\ell} / p(\{\tau_\ell, \dots, \tau_k\})$, which is the Plackett–Luce law [15, 22]. Choosing L from p, we obtain the following.

COROLLARY 4.9 (Successive sampling on forests). *Let f be a B-parent map with root blocks D_u, let $p_v > 0$ for $v \in B^c$, write $p(J) := \sum_{v \in J} p_v$ for $J \subseteq B^c$, and for a growth order τ let B_ℓ be its bases. Then, for every $r \in B$,*

$$\sum_{\tau \in \mathcal{S}_B(f):\ f(\tau_1) = r} \ \prod_{1 \le \ell \le k-1:\ f(\tau_{\ell+1}) \neq \tau_\ell} \frac{p_{\tau_\ell}}{p(B^c_{\ell-1})} = \frac{p(D_r)}{p(B^c)}. \tag{4.4}$$

In particular, with $B = \{r\}$ and p uniform on B^c,

$$\sum_{\tau \in \mathcal{S}_{\{r\}}(f)} \prod_{1 \le \ell \le k-1:\ \tau_\ell = f(\tau_{\ell+1})} (k - \ell + 1) = k! \,. \tag{4.5}$$

The key step in the proof is to take $L := I - \mathbf{1}p^\top$ in Theorem 4.5; we defer the details to section B. Note that the index set in (4.5) is the complement of that in (4.4), since $\prod_{\ell=1}^{k-1}(k - \ell + 1) = k!$.

Remark 4.10 (Linear extensions). In other words, (4.5) weights each linear extension of a rooted tree by a positional statistic, and the weights sum to $k!$ for every tree. For example, a star has $k!$ linear extensions of weight one, and a path has one, of weight $k!$.

Remark 4.11. The order-mass identity (Theorem 4.5) also extends from root indicators to real aggregations with unit row sums. We present the generalization in Appendix C.

5. Proof of the order-mass identity. We prove the identity by induction on the number of nonroot vertices. The main operation is to move the first discovered vertex w into the root set. The remaining order then grows a forest rooted at $B \cup \{w\}$, whose order masses are known by induction. To finish the step, we must express those masses through the cofactors on the original root set B.

This operation explains the role of multiple roots in the proof. Even when the original forest is a tree, moving its first vertex creates a second root. We therefore establish the identity for all root sets before using its single-root specialization.

The key step is a lemma about one base U and the base $U' := U \cup \{w\}$ obtained by adjoining one vertex. It rests on two facts. First, adjoining w changes every harmonic extension by a multiple of the one fixed vector $\gamma^U(e_w)$:

$$\gamma^{U'}(m) = \gamma^U(m) + \big(\gamma^{U'}(m)\big)_w \, \gamma^U(e_w) \qquad (m \in \mathbb{R}^{U^c},\ m_w = 0), \tag{5.1}$$

where on the left, m is read as a vector on $(U')^c$. Indeed, the difference of the two sides is annihilated by the rows U of L and vanishes off U; hence it is $\mathbf{0}$ by uniqueness. Second, Lemma 2.1(b) evaluates a rank-one change through the cofactor row of a matrix with zero row sums. These two facts give the transfer relation below.

5.1. The one-vertex transfer lemma. For $U \subsetneq V$ with $\det L[U] \neq 0$ and a matrix M with rows indexed by U^c and columns by U, we write

$$N(U, M) := I - \Gamma^U M, \qquad N(U, M)_{\cdot u} = e_u - \Gamma^U M_{\cdot u} \quad (u \in U). \tag{5.2}$$

When $M\mathbf{1} = \mathbf{1}$ and $L\mathbf{1} = \mathbf{0}$, the rows of $N(U, M)$ sum to zero, since $N(U, M)\mathbf{1} = \mathbf{1} - \Gamma^U \mathbf{1} = \mathbf{0}$. Note that the right side of (4.2) is a cofactor of $N(B, M_f)$.

The transfer lemma compares two aggregations. After the move, M' has rows indexed by $(U')^c$, columns indexed by U', and unit row sums. We recover the aggregation before the move by assigning the new root w back to a root $r \in U$. Specifically, the *fold* of M' into r is the matrix M with rows U^c and columns U obtained by adding the column w of M' to the column r, and inserting $e_r^\top$ as the row of w:

$$M_{w\cdot} := e_r^\top, \qquad M_{v\cdot} := M'_{v,U} + M'_{vw}\, e_r^\top \quad (v \in (U')^c). \tag{5.3}$$

Clearly, its rows sum to one. For ease of notation, we read every column of M' as a vector on U^c with the entry 0 at w. Then (5.3) says that, for $u \in U$,

$$M_{\cdot u} = M'_{\cdot u} + [u = r]\,\big(M'_{\cdot w} + e_w\big). \tag{5.4}$$

The following lemma relates the principal cofactors before and after the move. Figure 2(b) and (c) illustrate the move and the fold.

LEMMA 5.1 (One-vertex transfer). *Let $L \in \mathbb{R}^{V\times V}$ satisfy $L\mathbf{1} = \mathbf{0}$; let $U \subsetneq V$ be nonempty with $|U| \le n-2$, $w \in U^c$, $U' := U \cup \{w\}$, $\det L[U] \neq 0$ and $\det L[U'] \neq 0$. Let M' have rows $(U')^c$, columns U' and unit row sums, let $r \in U$, and let M be the fold (5.3) of M' into r. Let $d' \in \mathbb{R}^{U'}$ and $d \in \mathbb{R}^{U}$ be the vectors of principal cofactors of $N(U', M')$ and $N(U, M)$. Then*

$$\sum_{u\in U'} \big(\gamma^{U}(e_w)\big)_u \, d'_u = \sum_{u\in U} \big(\gamma^{U}(M'_{\cdot w} + e_w)\big)_u \, d_u. \tag{5.5}$$

The proof evaluates one auxiliary determinant using each base. Applying Lemma 2.1(b) to the enlarged base gives the left side of (5.5); applying it to the original base gives the right side.

Proof of Lemma 5.1. We write $N' := N(U', M')$ and $N := N(U, M)$; both have zero row sums. We also write

$$G := \Big[\gamma^{U}(M'_{\cdot u})\big|_{U'}\Big]_{u\in U'} \in \mathbb{R}^{U'\times U'}, \qquad a := \gamma^{U}(e_w)\big|_{U'} \in \mathbb{R}^{U'}.$$

We first express the change in N' as a rank-one term. Fix $u \in U'$. By (5.2), the column u of N' is $e_u - \gamma^{U'}(M'_{\cdot u})\big|_{U'}$. Since $M'_{\cdot u}$ vanishes at w, (5.1) applies to $m := M'_{\cdot u}$, and its coefficient is $\big(\gamma^{U'}(M'_{\cdot u})\big)_w = (\Gamma^{U'} M')_{wu}$. Restricting (5.1) to U', we obtain

$$\gamma^{U'}(M'_{\cdot u})\big|_{U'} = G_{\cdot u} + (\Gamma^{U'} M')_{wu}\, a, \qquad \text{hence} \qquad N'_{\cdot u} = e_u - G_{\cdot u} - (\Gamma^{U'} M')_{wu}\, a.$$

Collecting the columns $u \in U'$, we have

$$N' = (I - G) - a\,(\Gamma^{U'} M')_{w\cdot}. \tag{5.6}$$

The determinant calculation uses three observations. (i) The row $(\Gamma^{U'} M')_{w\cdot}$ sums to one, since $M'\mathbf{1} = \mathbf{1}$ and $\Gamma^{U'}\mathbf{1} = \mathbf{1}$. (ii) The row w of G vanishes, since $(\gamma^{U}(M'_{\cdot u}))_w = (M'_{\cdot u})_w = 0$. (iii) For $u \in U$, (5.4) and the linearity of γ^{U} give $G_{\cdot u}|_U = \Gamma^{U} M_{\cdot u} - [u = r]\,\Gamma^{U}(M'_{\cdot w} + e_w)$.

We now evaluate $\det(I - G)$ in two ways.

Evaluation using the enlarged base. By (5.6), $I - G = N' + a\,(\Gamma^{U'} M')_{w\cdot}$. Applying Lemma 2.1(b) to N', together with (i), we have

$$\det(I - G) = a^{\top} d' = \sum_{u\in U'} \big(\gamma^{U}(e_w)\big)_u \, d'_u. \tag{5.7}$$

Evaluation using the original base. By (ii), the row w of $I - G$ is $e_w^{\top}$; hence $\det(I - G) = \det((I - G)[U])$. By (iii) and (5.2), the column $u \in U$ of $(I - G)[U]$ is

$$e_u - \Gamma^{U} M_{\cdot u} + [u = r]\,\Gamma^{U}(M'_{\cdot w} + e_w) = N_{\cdot u} + [u = r]\,\Gamma^{U}(M'_{\cdot w} + e_w),$$

that is,

$$(I - G)[U] = N + \Gamma^{U}(M'_{\cdot w} + e_w)\, e_r^{\top}. \tag{5.8}$$

Since $N\mathbf{1} = \mathbf{0}$, Lemma 2.1(b) gives

$$\det\big((I - G)[U]\big) = \big(\Gamma^{U}(M'_{\cdot w} + e_w)\big)^{\top} d = \sum_{u\in U} \big(\gamma^{U}(M'_{\cdot w} + e_w)\big)_u \, d_u.$$

Combining the two evaluations of $\det(I - G)$ gives (5.5). This completes the proof. ∎

5.2. Proof of the theorem. We now apply the transfer lemma to a forest. The induction step has three parts: group growth orders by their first vertex, apply the identity to the shortened orders, and fold the new root back into its parent. Summing over the children of a root then gives its cofactor. We first treat root sets with at least two elements and finish with the single-root case.

Proof of Theorem 4.5. Step 1: root sets with at least two elements. We induct on k, for all root sets with at least two elements and all parent maps at once, L fixed. Write $N := N(B, M_f)$ and $d_u := \det(N[B \setminus \{u\}])$, so that $d^\top N = \mathbf{0}^\top$ by Lemma 2.1(a); the right side of (4.2) is d_r.

For the base case $k = 1$, $B^c = \{w\}$ with $f(w) =: u \in B$; the unique growth order (w) has the empty product, so the left side is $[r = u]$. By (2.2), $\Gamma^B = \mathbf{1}$ and $\Gamma^B M_f = \mathbf{1}e_u^\top$, so $N = I - \mathbf{1}e_u^\top$ and (2.1) gives $d_r = 1 - \sum_{v \in B\setminus\{r\}} (e_u)_v = [r = u]$.

For the induction step $k \geq 2$, fix $r \in B$ and group the growth orders with $f(\tau_1) = r$ by their first vertex $w \in f^{-1}(r)$. We put $B' := B \cup \{w\}$ and let f' be the restriction of f to $B^c \setminus \{w\}$. Then f' is a B'-parent map whose bases are bases of f, so Theorem 4.1 holds for (B', f'); the map $\tau \mapsto (\tau_2, \dots, \tau_k)$ is a bijection from the growth orders of f starting at w onto $\mathcal{S}_{B'}(f')$; and $\omega_f(\tau) = (\gamma^B(e_w))_{f(\tau_2)}\, \omega_{f'}(\tau_2, \dots, \tau_k)$, where the last order weight is taken with respect to the root set B'. Grouping the shortened orders by the root $u := f(\tau_2) \in B'$ at which they enter, with the factor $(\gamma^B(e_w))_u = \Gamma^B_{uw}$ for $u \in B$ and $(\gamma^B(e_w))_w = 1$ for $u = w$, and applying the induction hypothesis to (B', f'), we obtain

$$g_r(B, f) = \sum_{w \in f^{-1}(r)} \Big(\sum_{u \in B} \Gamma^B_{uw} \det\big(N'_w[B' \setminus \{u\}]\big) + \det\big(N'_w[B' \setminus \{w\}]\big) \Big), \tag{5.9}$$

where $N'_w := N(B', M_{f'})$.

To transfer these cofactors back to B, fix w. The root blocks of (B', f') are D_u for $u \in B \setminus \{r\}$, $D_r \setminus D_w$ for r, and $D_w \setminus \{w\}$ for w. Hence the fold of $M_{f'}$ into r is M_f. Moreover, the column w of $M_{f'}$ is the indicator of $D_w \setminus \{w\}$, so $M_{f'\cdot w} + e_w$ is the indicator $\mathbf{1}_{D_w}$ of D_w. Lemma 5.1 with $U := B$, $U' = B'$ and $M' := M_{f'}$ applies, since $|B| \leq n - 2$ because $k \geq 2$ and both B and B' are bases of f. Its left side is the inner sum of (5.9), and its right side is $\sum_{u \in B} (\Gamma^B \mathbf{1}_{D_w})_u d_u$. By Lemma 2.6 the blocks D_w of the children of r partition D_r, and $\Gamma^B \mathbf{1}_{D_r} = (\Gamma^B M_f)_{\cdot r} = e_r - N_{\cdot r}$. Summing over $w \in f^{-1}(r)$, we therefore have

$$g_r(B, f) = \sum_{u \in B} (e_r - N_{\cdot r})_u d_u = d_r - (d^\top N)_r = d_r,$$

where the last equality holds since $d^\top N = \mathbf{0}^\top$.

Step 2: a single root. Let $B = \{r\}$. If $k = 1$, the unique growth order has the empty product 1. Now let $k \geq 2$, so that $n \geq 3$ and $|B| \leq n - 2$; every growth order then starts at a child w of r. We fix w, put $B' := \{r, w\}$, and let f' be the restriction of f to $B^c \setminus \{w\}$. As in the case $|B| \geq 2$, (B', f') satisfies Theorem 4.1, the shortened orders are its growth orders, and the case $|B| = 2$ applies to (B', f'). Hence the sum of $\omega_f(\tau)$ over the growth orders starting at w equals the left side of (5.5) for $U := \{r\}$ and $M' := M_{f'}$, which is $\det(I - G)$ by (5.7); there Lemma 2.1(b) is applied only to the 2×2 matrix $N(B', M_{f'})$. Therefore, it remains to evaluate $\det(I - G)$ directly. By fact (ii), $\det(I - G)$ is the entry of the 1×1 matrix $(I - G)[\{r\}]$, which by (5.8) is $N_{rr} + (\Gamma^{\{r\}} \mathbf{1}_{D_w})_r$. The 1×1 matrix $N = N(\{r\}, M_f)$ has zero row sums, so $N_{rr} = 0$. Hence the sum of $\omega_f(\tau)$ over the growth orders starting at w is $\Gamma^{\{r\}} \mathbf{1}_{D_w}$. The blocks

of the children of r partition $D_r = B^c$ by Lemma 2.6. Summing over w and using $\Gamma^{\{r\}}\mathbf{1} = 1$, we conclude that

$$\sum_{\tau \in \mathcal{S}_{\{r\}}(f)} \omega_f(\tau) = \Gamma^{\{r\}}\mathbf{1} = 1.$$

This completes the proof. ∎

Remark 5.2 (Where the hypotheses act). The zero-row-sum condition supplies the normalization (2.2). It ensures that the aggregated matrices have zero row sums, so Lemma 2.1 applies, and that $(\Gamma^{U'}M')_{w\cdot}$ sums to one in (5.6). It also gives $d^\top N = \mathbf{0}^\top$, which cancels the final term in the induction step. Nonsingularity is used to define the harmonic extensions and to compare them on successive bases.

6. Consequences for the covering walk. We return to the Markov chain and take L as in (3.1). Throughout this section, P is irreducible and stochastic, π is its stationary distribution, and λ is the cofactor vector (3.4). The order-mass identity now supplies the normalization missing from the order decomposition. We first obtain the known tree law, then the conditional order law, and finally the stopped-forest law. Individual transition probabilities may be zero; Theorem A.4 illustrates all three laws on a chain with one-way support.

6.1. The Aldous–Broder theorem. The single-root identity evaluates the sum in (3.9). Substituting it into the order decomposition gives the Aldous–Broder theorem in the generality of [11].

THEOREM 6.1 (Aldous–Broder for general chains [11, Thm. 3.4]). *Let $r \in V$ and let F be the first-entrance tree of the covering walk started at r. For every $F_0 \in \mathcal{F}_r$ with $\{r\}$-parent map f,*

$$\mathbb{P}(F = F_0) = \frac{w(F_0)}{(\mathbf{1}^\top\lambda)\prod_{v\in V}\pi_v} = \frac{w(F_0)}{\sum_{F'\in\mathcal{F}_r} w(F')}, \qquad w(F_0) = \prod_{v\neq r} \pi_{f(v)}P_{f(v)v}.$$

Proof. By Lemma 3.3, Theorem 4.1 holds for every arborescence. Summing (3.6) over $\tau \in \mathcal{S}_{\{r\}}(f)$ and applying the single-root case of Theorem 4.5, we obtain the first form. Since the probabilities sum to one over $F_0 \in \mathcal{F}_r$, their common denominator equals the sum of the numerators, which gives the second form. ∎

The normalizing constant also yields the arborescence form of the Markov chain tree theorem for P^* by summing over F_0. See [12] for a direct linear-algebra proof of that theorem. Theorem A.2 evaluates the tree law on the four-state chain.

6.2. The order in which the tree is discovered. For a fixed tree of positive weight, the order weights sum to one. Dividing the joint forest-and-order probability by the tree probability identifies these weights as the conditional distribution of the discovery order. Thus the order decomposition describes both how likely the tree is and how it is discovered.

THEOREM 6.2 (Discovery order given the tree). *Let $r \in V$, let F and $\tau(X)$ be the first-entrance tree and discovery order of the covering walk started at r, and let $F_0 \in \mathcal{F}_r$ have parent map f and positive weight $w(F_0) > 0$. Then*

$$\mathbb{P}\big(\tau(X) = \tau \mid F = F_0\big) = \omega_f(\tau) \qquad (\tau \in \mathcal{S}_{\{r\}}(f)),$$

and $\mathbb{P}(\tau(X) = \tau \mid F = F_0) = 0$ for every other ordering τ.

Proof. By Theorem 6.1, we have $\mathbb{P}(F = F_0) > 0$. Dividing (3.6) by the first form of Theorem 6.1 cancels the tree weight and the constant. For the orderings that are not growth orders, the claim follows from Proposition 3.2. ∎

This conditional law follows from the order decomposition together with any proof of the tree law. The role of the order-mass identity in our route is to establish the normalization directly, before the tree law is known.

Example 6.3 (A nonuniform law). The law need not be uniform on the growth orders. For example, for simple random walk on the complete graph, the conditional probability of τ is the summand of (4.5) divided by $(n-1)!$, since the matrix L is then a multiple of the Laplacian of the complete graph, and harmonic extensions are unchanged by scaling.

6.3. The stopped covering walk. With several roots, the order masses need not equal one. To obtain the stopped-forest law, we instead sum the weights of the ways to complete a fixed forest into a tree rooted at the seed. We first note that the forest F_B is the restriction of the tree F of Theorem 6.1 to the edges into B^c, since each first-entrance edge into B^c is determined at time $T_v \le T_B$. Conversely, given a forest $F_0 \in \mathcal{F}_B$ and $r \in B$, an arborescence $F' \in \mathcal{F}_r$ with $F' \supseteq F_0$ consists of F_0 together with one edge into each root $r' \in B \setminus \{r\}$; we refer to such an arborescence as a *completion* of F_0. We define the *completion weight* of F_0 at r by summing the products of the added-edge weights,

$$\varphi_r(F_0) := \sum_{F' \in \mathcal{F}_r:\ F' \supseteq F_0} \ \prod_{u \in B \setminus \{r\}} \pi_{f_{F'}(u)} P_{f_{F'}(u)\,u}, \tag{6.1}$$

where $f_{F'}$ is the parent map of F'. Since the edges of F_0 and the added edges are disjoint, $\sum_{F' \in \mathcal{F}_r:\ F' \supseteq F_0} w(F') = w(F_0)\,\varphi_r(F_0)$. This quantity is the weighted analogue of the number of spanning trees containing a fixed forest. Such tree marginals can be computed by contraction [17, §4.2], which also appears in the two-stage samplers of [19, §5]. Here a completion adds one edge into each root other than r; the edges into the nonroot vertices remain those of F_0.

THEOREM 6.4 (Law of the stopped covering walk). *Let $B \subsetneq V$ be nonempty with seed $r \in B$, and let F_B be the first-entrance forest of the stopped covering walk. For every $F_0 \in \mathcal{F}_B$ with parent map f,*

$$\mathbb{P}(F_B = F_0) = \frac{\sum_{F' \in \mathcal{F}_r:\ F' \supseteq F_0} w(F')}{\sum_{F' \in \mathcal{F}_r} w(F')} = \frac{w(F_0)\,\varphi_r(F_0)}{(\mathbf{1}^\top \lambda) \prod_{v \in V} \pi_v}, \tag{6.2}$$

and the completion weight is

$$\varphi_r(F_0) = \det L[B] \cdot \prod_{u \in B \setminus \{r\}} \pi_u \cdot g(B,f)^\top L[B]^{-1} e_r. \tag{6.3}$$

Proof. We first prove (6.2). Since F_B is the restriction of F, summing Theorem 6.1 over the completions of F_0 gives (6.2). We now prove (6.3). Summing (3.5) over $\tau \in \mathcal{S}_B(f)$, grouped by $u := f(\tau_1)$, produces by (4.1) the factor $\sum_{u \in B} g_u(B,f)(L[B]^{-1})_{ur} = g(B,f)^\top L[B]^{-1} e_r$, so that

$$\mathbb{P}(F_B = F_0) = \frac{\det L[B]\ w(F_0)\ g(B,f)^\top L[B]^{-1} e_r}{\pi_r \big(\prod_{v \in B^c} \pi_v \big) \mathbf{1}^\top \lambda}.$$

Comparing with (6.2) and canceling $w(F_0)$ gives (6.3) when $w(F_0) > 0$. For a forest of zero weight, we replace P by $P_\varepsilon := (1-\varepsilon)P + \varepsilon \mathbf{1}\mathbf{1}^\top/n$, $0 < \varepsilon < 1$, for which every forest has positive weight. By Lemma 3.4, the stationary distribution of P_ε is a continuous function of the principal cofactors of $I - P_\varepsilon$ and converges to π, and the minors entering the harmonic extensions and $L[B]^{-1}$ remain nonzero at the limit by Lemma 3.3. Therefore, both sides of (6.3) are continuous in ε, and letting $\varepsilon \downarrow 0$ proves (6.3) for every forest. This completes the proof. ∎

The law (6.2) follows from Theorem 6.1 alone; the content of Theorem 6.4 is the evaluation (6.3) of the completion weight through the order masses. For $B = \{r\}$, the only completion is F_0 itself with no added edge, $\varphi_r(F_0) = 1$, and (6.2) reduces to Theorem 6.1. In general, $\varphi_r(F_B)$ carries the whole dependence of the law on the shape of the forest beyond its weight. The cofactor representation gives a direct way to compute it. Since $L[B] + L[B, B^c]M_f = L[B]\,(I - \Gamma^B M_f)$ and $\operatorname{adj}(L[B]) = \det(L[B])\, L[B]^{-1}$, the product rule $\operatorname{adj}(XY) = \operatorname{adj}(Y)\operatorname{adj}(X)$ for adjugates and Lemma 2.1(a) give

$$\varphi_r(F_0) = \Big(\prod_{u \in B\setminus\{r\}} \pi_u \Big) \det\big((L[B] + L[B, B^c]M_f)[B \setminus \{r\}] \big), \tag{6.4}$$

which requires one determinant of size $|B| - 1$ per forest. We tabulate the resulting law for the four-state example in section A. Since $\varphi_r(F_0)$ depends on F_0 only through the block indicator M_f, forests with the same partition of B^c into root blocks share one evaluation.

Remark 6.5 (Computing the completion weight). Formula (6.4) is an analytic evaluation, and we do not claim a speed advantage over Wilson's algorithm or other samplers. Its cost is as follows, with $b := |B|$ and $k := |B^c|$. For a fixed chain and root set, we compute π, by one linear solve of size n, and the blocks $L[B]$ and $L[B, B^c]$ once. For each forest, the matrix $L[B] + L[B, B^c]M_f$ is assembled in $O(bk)$ operations by adding the columns of $L[B, B^c]$ over the root blocks, and its principal minor at r costs $O(b^3)$, which is not small when b is close to n.

Conditioning the joint order decomposition on the stopped forest gives its discovery-order law. Unlike the single-root case, the law also depends on the seed through a column of $L[B]^{-1}$.

Corollary 6.6 (Discovery order given the stopped forest). *Under the assumptions of Theorem 6.4, let $F_0 \in \mathcal{F}_B$ have parent map f and $\mathbb{P}(F_B = F_0) > 0$. Then, for every $\tau \in \mathcal{S}_B(f)$,*

$$\mathbb{P}\big(\tau(X) = \tau \mid F_B = F_0\big) = \frac{\big(L[B]^{-1}\big)_{f(\tau_1),\, r}\, \omega_f(\tau)}{\sum_{u \in B} g_u(B, f)\big(L[B]^{-1}\big)_{ur}},$$

and $\mathbb{P}(\tau(X) = \tau \mid F_B = F_0) = 0$ for every other ordering τ of B^c. In particular, the growth orders entering at $u \in B$ carry the conditional mass

$$\frac{g_u(B, f)\big(L[B]^{-1}\big)_{ur}}{\sum_{u' \in B} g_{u'}(B, f)\big(L[B]^{-1}\big)_{u'r}}.$$

Proof. Dividing (3.5) by the expression of $\mathbb{P}(F_B = F_0)$ in the proof of Theorem 6.4 cancels the forest weight and the constant; the denominator is nonzero since $\mathbb{P}(F_B = F_0) > 0$. For the orderings that are not growth orders, the claim follows from Proposition 3.2. This completes the proof. ∎

For $B = \{r\}$, the factor $(L[B]^{-1})_{rr}$ cancels, and Corollary 6.6 reduces to Theorem 6.2.

Remark 6.7 (The contracted chain). In the chain instance, the matrix

$$Q := P^*[B] + P^*[B, B^c]M_f$$

is a transition matrix on B. Starting from a root, the reversed chain makes one step; we then record the root of the tree containing the destination. This construction gives a second evaluation of the completion weight.

To see it, consider an added edge into a root $u \neq r$. If its parent x lies in the tree of u', the sum of its weights over that tree is $\sum_x \pi_x P_{xu} = \pi_u Q_{uu'}$. Group the completions by the tree obtained after contracting each component of F_0 to its root. For each contracted tree, summing over the possible endpoints of its added edges gives the corresponding product of these flows. The matrix-tree theorem [27, 5], applied to $I - Q = L[B] + L[B, B^c]M_f$, therefore gives (6.4). In particular, $\varphi_r(F_0)/\prod_{u\in B\setminus\{r\}} \pi_u$ is the principal cofactor at r of $I - Q$. These cofactors form an unnormalized invariant measure of Q by Lemma 2.1(a). For a fixed seed, larger cofactors correspond to larger completion weights.

The two chains on the roots describe different operations. The matrix $\Gamma^B M_f$ of Theorem 4.7 records the tree entered when the reversed chain first exits B, after any number of steps inside B. The matrix Q records one step followed by aggregation. Their relation is $I - Q = L[B]\,(I - \Gamma^B M_f)$. The product rule for adjugates then gives the cofactor row of $I - Q$ as $\det L[B]\, g(B, f)^\top L[B]^{-1}$, in agreement with (6.3) and (6.4). Thus the order masses are the cofactors of $I - \Gamma^B M_f$, whereas the completion weights are the cofactors of $I - Q$ multiplied, at each root r, by $\prod_{u\in B\setminus\{r\}} \pi_u$.

Remark 6.8 (Relation to Wilson's algorithm). Wilson's algorithm run with P^* and the roots B absorbing samples forests F_B with law exactly proportional to $w(F_B)$ [29, 24]; see [3, Cor. 2.3] for a fixed root set and [18] for another proof. In comparison, Theorem 6.4 shows that the stopped covering walk samples this law tilted by the completion weight $\varphi_r(F_B)$. In Appendix D, we show that randomizing the seed cannot remove the tilt in general.

The next two remarks discuss when the tilt induced by $\varphi_r(F_B)$ is absent, and how to correct for it by importance weighting when it is present.

Remark 6.9 (When the tilt vanishes). By (6.2), the walk from the seed r samples exactly the law proportional to $w(F_B)$ if and only if the completion weight $\varphi_r(F_0)$ is the same for all forests F_0 of positive weight. In particular, this holds when every arc from B^c into B ends at r. To see this, note that in this case every completion of positive weight attaches the roots in $B \setminus \{r\}$ through arcs inside B, so $\varphi_r(F_0)$ does not depend on F_0. For a reversible chain, this condition means that every path between B^c and $B \setminus \{r\}$ passes through r. It can therefore hold, but requires a particular separation of the root set. In a nonreversible chain, it can hold without this undirected separation, as Theorem A.4 shows.

The completion weight also describes the scope of importance correction. Before using the stopped walk to estimate a functional under the weight-proportional law, one must determine which forests the walk can reach. The following remark separates this support question from the variability of the importance weights. We illustrate both effects in section F.

Remark 6.10 (Importance correction). Write $q_W(F_0) = w(F_0)/Z_W$ for the weight-proportional law we target, $q_A(F_0) = w(F_0)\varphi_r(F_0)/Z_A$ for the law (6.2) we

sample forests from, and $S_r := \{F_0 \in \mathcal{F}_B : \ w(F_0) > 0, \ \varphi_r(F_0) > 0\}$ for the support of q_A. Then, for every function h on $\mathcal{F}_B$,

$$\frac{\mathbb{E}_{q_A}\big[h(F_B)/\varphi_r(F_B)\big]}{\mathbb{E}_{q_A}\big[1/\varphi_r(F_B)\big]} = \mathbb{E}_{q_W}\big[h(F_B) \mid F_B \in S_r\big].$$

The ratio therefore estimates the target expectation conditional on the reachable forests. It recovers $\mathbb{E}_{q_W}[h]$ for every function h if and only if $q_W(S_r) = 1$. For a particular h, the conditional and unconditional expectations can agree despite incomplete support; a constant h is one example. Support loss also imposes a limit on distributional accuracy: any law $\widetilde{q}$ supported on S_r satisfies $\mathrm{TV}(\widetilde{q}, q_W) \geq 1 - q_W(S_r)$.

For independent draws from q_A, the self-normalized sample ratio is consistent for the conditional target. At finite sample sizes it is generally biased for that target, even with complete support. It is unbiased for the conditional target when h or φ_r is constant on S_r. Small positive completion weights can nevertheless produce variable estimates. Section F examines support loss, weight dispersion, and estimation separately.

7. Discussion. The order-mass identity evaluates a sum over growing domains through one cofactor on the root set. Its single-root specialization gives a direct proof of the Aldous–Broder law, while the several-root form evaluates the completion weight in the stopped-forest law. The same calculation identifies the conditional distribution of the discovery order. The sampling implication is explicit: stopping after the nonroot vertices have been covered generally changes the forest law, and inverse-completion weighting recovers only the target law on the reachable forests unless the support is complete.

Several questions remain. First, the conditional law in Theorem 6.2 gives a distribution on the growth orders of a fixed tree. Its later-position marginals and the cost of sampling from it remain to be analyzed. The weighted count in Corollary 4.9 also suggests studying order sums for other choices of the matrix.

Second, a bijective proof of Theorem 4.5 could explain how the growth orders combine into the forest sums of Theorem 4.8. The denominators $\prod_\ell \det L[B_{\ell-1}]$ depend on the entire order, so a weight-preserving correspondence would have to account for the changing domains. Connections with loop erasure, such as those in [8], may be useful for this question.

Third, the completion formula provides criteria for assessing other walk-based forest samplers. A useful sampler must reach the forests required by the target law; when reweighting is needed, small completion weights can lead to variable estimates. The numerical examples show these two effects separately. Designing samplers with complete support and stable importance weights, and determining their computational cost on graphs used in clustering and signal processing, remain problems for future work.

Acknowledgments. We thank Dr. Leo Duan for the useful discussion for this project.

Disclosure of the use of generative AI. Generative-AI tools assisted with language editing and coding. The author takes full responsibility for the mathematical statements, the correspondence between the paper and the formalization, the citations and the final manuscript.

Appendix A. Worked examples.

The examples illustrate three features of the results. A four-state chain makes the harmonic maps, order masses, and completion weights explicit. A chain with one-way transitions shows the roles of reversal and support. A signed matrix checks the identity beyond the probabilistic setting, where individual order weights can be negative.

Example A.1 (A four-state chain). Let $V = \{1,2,3,4\}$ and $L := I - \frac{1}{5}W^\top$ with

$$W = \begin{pmatrix} 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 1 & 1 \\ 2 & 1 & 1 & 1 \end{pmatrix}.$$

Every column of W sums to 5, so $L\mathbf{1} = \mathbf{0}$, and L is the matrix (3.1) of the doubly stochastic chain $P := W/5$. We take $B = \{1,2\}$, so that the minors required by Theorem 4.1 are among

$$\det L[\{1,2\}] = \tfrac{3}{5}, \quad \det L[\{1,2,3\}] = \det L[\{1,2,4\}] = \tfrac{36}{125}.$$

We have

$$\tilde{\Gamma}^B = \begin{pmatrix} \gamma^B(e_3) & \gamma^B(e_4) \end{pmatrix}, \quad \gamma^B(e_3) = (\tfrac{2}{5}, \tfrac{3}{5}, 1, 0)^\top, \quad \gamma^B(e_4) = (\tfrac{3}{5}, \tfrac{2}{5}, 0, 1)^\top.$$

We consider the three B-parent maps in Figure 3

$$f_1 : \ 3 \mapsto 1, \ 4 \mapsto 3; \qquad f_2 : \ 3 \mapsto 2, \ 4 \mapsto 3; \qquad f_3 : \ 3 \mapsto 1, \ 4 \mapsto 2,$$

with root blocks $(D_1, D_2) = (\{3,4\}, \emptyset)$, $(\emptyset, \{3,4\})$ and $(\{3\}, \{4\})$, respectively.

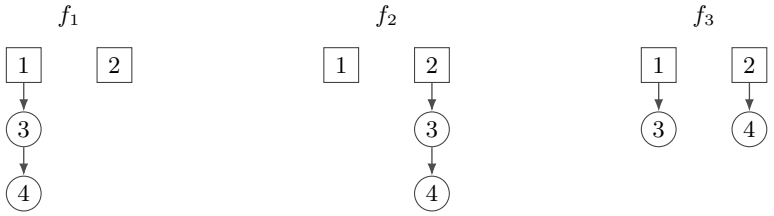


FIG. 3. *The forests f_1, f_2, f_3 of Example A.1, rooted at $B = \{1,2\}$.*

The matrices $I - \Gamma^B M_f$ are

$$\begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \qquad \begin{pmatrix} \frac{3}{5} & -\frac{3}{5} \\ -\frac{3}{5} & \frac{3}{5} \end{pmatrix},$$

with the vectors of principal cofactors $(1,0)^\top$, $(0,1)^\top$ and $(\frac{3}{5}, \frac{3}{5})^\top$, respectively. On the left of (4.2), f_1 has the single growth order $(3,4)$, entering at 1 with the factor $(\gamma^B(e_3))_3 = 1$, and f_2 likewise enters at 2. On the other hand, f_3 has $(3,4)$ entering at 1 with the factor $(\gamma^B(e_3))_2 = \frac{3}{5}$, and $(4,3)$ entering at 2 with $(\gamma^B(e_4))_1 = \frac{3}{5}$. These agree with the principal cofactors above.

In addition, consider the single root 1 and the star $f_\star : 2,3,4 \mapsto 1$. Every ordering of $\{2,3,4\}$ is a growth order, and the six order weights

$$\tfrac{1}{10}, \ \tfrac{3}{20}, \ \tfrac{1}{10}, \ \tfrac{3}{20}, \ \tfrac{1}{4}, \ \tfrac{1}{4} \qquad \text{for } (2,3,4), (2,4,3), (3,2,4), (3,4,2), (4,2,3), (4,3,2)$$

sum to 1, as the single-root identity requires.

Example A.2 (Theorem A.1 continued). The principal cofactors of $I - P$ are $\lambda_v = \frac{36}{125}$ for every v, so $\mathbf{1}^\top \lambda = \frac{144}{125}$. With the single root 1, the star $f_\star$ has tree weight

$$w(F_\star) = \prod_{v \neq 1} \pi_{f(v)} P_{f(v)v} = \prod_{v \neq 1} \pi_1 P_{1v} = \frac{1}{4000},$$

and hence

$$\mathbb{P}(F = F_\star) = \frac{1/4000}{(144/125)(1/256)} = \frac{1}{18}.$$

The six order weights of Example A.1 are the proportions in which the six growth orders contribute to this probability. In addition, one can verify that the 16 arborescences rooted at 1 have total weight $\frac{9}{2000}$, as the second form of Theorem 6.1 requires.

Example A.3 (Theorem A.1 continued). We now return to the chain of Example A.1 with $B = \{1, 2\}$, for which π is uniform and $\det L[B] = \frac{3}{5}$. With the order masses $(1, 0)$, $(0, 1)$ and $(\frac{3}{5}, \frac{3}{5})$ of f_1, f_2, f_3, (6.3) and (6.2) give the completion weights and probabilities of three of the eight forests rooted at B in Table 1.

TABLE 1

Completion weights and laws of F_B for the forests f_1, f_2, f_3 of Figure 3; $\mathbb{P}_r$ is the law of F_B under the seed r.

forest F_0	$w(F_0)$	$\varphi_1(F_0)$	$\varphi_2(F_0)$	$\mathbb{P}_1(F_B = F_0)$	$\mathbb{P}_2(F_B = F_0)$
f_1	1/400	1/5	1/20	1/9	1/36
f_2	1/200	1/20	1/5	1/18	2/9
f_3	1/400	3/20	3/20	1/12	1/12

Note that the first two forests have weights in ratio 1:2, but their seed-1 probabilities are in ratio 2:1: the walk from 1 prefers the forest whose large tree hangs from its own seed.

Example A.4 (A chain with one-way support). The doubly stochastic chain of Example A.1 is convenient for computation, but it hides the role of the reversal. We now consider $V = \{1, 2, 3, 4\}$ and the chain of Figure 4,

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & 0 & 0 & \frac{2}{3} \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad P^* = \begin{pmatrix} 0 & 0 & \frac{1}{4} & \frac{3}{4} \\ 1 & 0 & 0 & 0 \\ \frac{2}{3} & \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & \frac{2}{3} & 0 \end{pmatrix},$$

with the stationary distribution $\pi = (\frac{1}{3}, \frac{1}{6}, \frac{1}{4}, \frac{1}{4})^\top$.

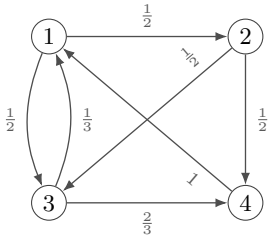


FIG. 4. *The transitions of the chain of Example A.4. Only the arc between 1 and 3 is used in both directions; every other arc is one-way.*

The chain is irreducible, since $1 \to 2 \to 4 \to 1$ and $1 \to 3 \to 1$ are cycles through every state. However, nine of the sixteen transitions vanish, the arcs $1 \to 2$, $2 \to 3$, $2 \to 4$, $3 \to 4$ and $4 \to 1$ are one-way, and $\pi_u P_{uv} \neq \pi_v P_{vu}$ for every pair $u \neq v$. In particular, P is not positive on the edges of any undirected graph, so the hypothesis of [9] fails. Nevertheless, the results of section 6 require only irreducibility, and we now check each of them. All the numbers below are exact, and we computed the laws of the walk independently by first-step analysis on the pairs (visited set, current state), using only the forward chain.

The tree law. Start the covering walk at $r = 1$. Of the 16 arborescences rooted at 1, only four have positive weight, and their total weight is $\frac{1}{96}$. Table 2 lists these four trees with their probabilities from Theorem 6.1 and the conditional law of their growth orders from Theorem 6.2; the other twelve trees have probability zero.

TABLE 2

The four trees of positive weight rooted at 1 *in Theorem A.4, written as* $(f(2), f(3), f(4))$*, with their weights, their probabilities under the covering walk from* 1*, and the order weights of their growth orders.*

tree	$w(F_0)$	$\mathbb{P}(F = F_0)$	growth orders τ and order weights $\omega_f(\tau)$
$(1,1,3)$	$1/216$	$4/9$	$(2,3,4)$: 0, $(3,2,4)$: $1/10$, $(3,4,2)$: $9/10$
$(1,1,2)$	$1/432$	$2/9$	$(2,3,4)$: 0, $(2,4,3)$: $3/4$, $(3,2,4)$: $1/4$
$(1,2,3)$	$1/432$	$2/9$	$(2,3,4)$: 1
$(1,2,2)$	$1/864$	$1/9$	$(2,3,4)$: $1/4$, $(2,4,3)$: $3/4$

The probabilities and the order weights both agree with the first-step analysis. Every tree of positive weight has $f(2) = 1$, since the only arc into 2 is from 1. In contrast, the law proportional to $\prod_{v \neq 1} P_{vf(v)}$, which Wilson's algorithm samples when run with P in place of P^*, is supported on the trees with $f(2) \in \{3, 4\}$, since 2 can only move to 3 or 4. The two laws are therefore mutually singular.

Order weights equal to zero. The growth order $(2, 3, 4)$ of the first two trees has weight zero. Its first factor is $(\gamma^{\{1\}}(e_2))_1 = P^*_{12} = 0$. Correspondingly, once the walk has entered 2, it cannot return to 1 without visiting 3 or 4, so it never discovers 2 and then 3 from 1. The order weights of each of the 16 trees still sum to one, as Theorem 4.5 requires, including the twelve trees of weight zero.

The stopped covering walk. Take $B = \{1, 2\}$. Then $L[B] = \left(\begin{smallmatrix} 1 & 0 \\ -1 & 1 \end{smallmatrix}\right)$, so $\det L[B] = 1$, and (6.3) gives $\varphi_1(F_0) = \pi_2\big(g_1(B, f) + g_2(B, f)\big)$ and $\varphi_2(F_0) = \pi_1\, g_2(B, f)$. Table 3 lists the eight forests rooted at B with their order masses, completion weights and probabilities under the two seeds. Both probability columns agree with (6.2) and with the first-step analysis.

The last four forests have weight zero, since $P_{14} = P_{43} = 0$, but their order masses are well defined and satisfy Theorem 4.5: the identity is a statement about L, not about the weight of the forest.

The two seeds behave very differently. Every order mass vector sums to one here, so $\varphi_1(F_0) = \pi_2 = \frac{1}{6}$ for every F_0, and hence $\mathbb{P}_1(F_B = F_0) \propto w(F_0)$; that is, the walk from 1 samples exactly the law proportional to forest weight $w(F_0)$. This is an instance of Theorem 6.9: every completion at 1 whose added edge has positive weight adds the edge $1 \to 2$, the only arc into 2, so that $\varphi_1(F_0) = \pi_1 P_{12} = \frac{1}{6}$ for every F_0. On the other hand, the walk using 2 as the seed is tilted by φ_2 and never returns the forest $(1, 3)$, the forest of largest weight. Indeed, its first step enters 3 or 4 from 2, and the completion weight φ_2 of $(1, 3)$ vanishes, since no vertex of the tree of 2 has an arc into

TABLE 3

The eight forests rooted at $B = \{1,2\}$ in Theorem A.4, written as $(f(3), f(4))$, with their weights, order masses, completion weights at the seed 2, and laws $\mathbb{P}_r$ of F_B under the seeds $r = 1, 2$. The completion weight at the seed 1 equals $\frac{1}{6}$ for every forest.

forest F_0	$w(F_0)$	$g(B,f)$	$\varphi_2(F_0)$	$\mathbb{P}_1(F_B = F_0)$	$\mathbb{P}_2(F_B = F_0)$
$(1,3)$	$1/36$	$(1,0)$	0	$4/9$	0
$(1,2)$	$1/72$	$(1/4,\ 3/4)$	$1/4$	$2/9$	$1/3$
$(2,3)$	$1/72$	$(0,1)$	$1/3$	$2/9$	$4/9$
$(2,2)$	$1/144$	$(0,1)$	$1/3$	$1/9$	$2/9$
$(1,1)$	0	$(1,0)$	0	0	0
$(2,1)$	0	$(3/4,\ 1/4)$	$1/12$	0	0
$(4,1)$	0	$(1,0)$	0	0	0
$(4,2)$	0	$(0,1)$	$1/3$	0	0

1. Since $(1,3)$ has probability $\frac{4}{9}$ under the law proportional to the forest weight, the target mass lost from the seed 2 is $\frac{4}{9}$, and every reweighting of $\mathbb{P}_2$ is at total variation distance at least $\frac{4}{9}$ from that law as discussed in Remark 6.10. Finally, (6.3) also holds for the forests of weight zero, as the proof of Theorem 6.4 asserts by continuity. For example, the forest $(2,1)$ has a single completion rooted at 2 whose added edge has positive weight, the edge $3 \to 1$, so that $\varphi_2(F_0) = \pi_3 P_{31} = \frac{1}{12} = \pi_1\, g_2(B,f)$, although the completed tree has weight zero.

Example A.5 (A matrix with entries of both signs). To show the generality of the order-mass identity in Theorem 4.5, we now consider a real matrix L that no Markov chain produces, so that some entries of the harmonic extensions are negative and some exceed one. We verified every value below in exact arithmetic. Let $V = \{1, \dots, 5\}$ and

$$L = \begin{pmatrix} 3 & -1 & 1 & -2 & -1 \\ -1 & 3 & -2 & 1 & -1 \\ 1 & -2 & 3 & -1 & -1 \\ -1 & -1 & -1 & 3 & 0 \\ -1 & 1 & -1 & -1 & 2 \end{pmatrix},$$

whose rows sum to zero and whose off-diagonal entries $L_{13}, L_{24}, L_{31}, L_{52}$ are positive. We take $B = \{1,2\}$ and $f : 3 \mapsto 1,\ 4 \mapsto 3,\ 5 \mapsto 2$, with root blocks $D_1 = \{3,4\}$ and $D_2 = \{5\}$, which is the forest of Figure 1. The three growth orders $(3,4,5)$, $(3,5,4)$, $(5,3,4)$ have the proper bases $\{1,2\}$, $\{1,2,3\}$, $\{1,2,5\}$, $\{1,2,3,4\}$, $\{1,2,3,5\}$, on which the minors of L are 8, 13, 16, 31 and 23, respectively. The harmonic extension map $\tilde{\Gamma}^B$ has the first two rows as

$$\Gamma^B = \begin{pmatrix} -\frac{1}{8} & \frac{5}{8} & \frac{1}{2} \\ \frac{5}{8} & -\frac{1}{8} & \frac{1}{2} \end{pmatrix},$$

and the aggregated matrix is

$$I - \Gamma^B M_f = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix},$$

with both principal cofactors equal to $\frac{1}{2}$. In the base $\{1,2,3\}$, $\Gamma_{\cdot 4}^{\{1,2,3\}} = (\frac{8}{13}, -\frac{1}{13}, \frac{1}{13})^\top$ and $\Gamma_{\cdot 5}^{\{1,2,3\}} = (\frac{5}{13}, \frac{14}{13}, \frac{12}{13})^\top$. The order weights are

$$\omega_f(3,4,5) = 1 \cdot \big(-\tfrac{1}{13}\big) = -\tfrac{1}{13}, \qquad \omega_f(3,5,4) = \tfrac{5}{8} \cdot \tfrac{12}{13} = \tfrac{15}{26}, \qquad \omega_f(5,3,4) = \tfrac{1}{2} \cdot 1 = \tfrac{1}{2}.$$

The first two orders enter at the root 1 and the third at 2, so the order masses are $-\frac{1}{13}+\frac{15}{26}=\frac{1}{2}$ at $r=1$ and $\frac{1}{2}$ at $r=2$, as Theorem 4.5 asserts. In particular, the check at $r=1$ balances a negative product from the base $\{1,2,3\}$ against a product of entries from two different bases.

Appendix B. Proofs of the auxiliary results.

We collect here the proofs of Lemmas 2.1, 2.6 and 3.3 and of Corollary 4.9, deferred from the main text.

Proof of Lemma 2.1. (a) Since $A\mathbf{1}=\mathbf{0}$, we have $\det A=0$, and thus $A\operatorname{adj}(A)=\mathbf{0}$ by (2.1); that is, every column of $\operatorname{adj}(A)$ lies in $\ker A$. If $\operatorname{rank} A=|S|-1$, then $\ker A=\mathbb{R}\mathbf{1}$, and each column is constant. If $\operatorname{rank} A\le|S|-2$, then every minor of order $|S|-1$ vanishes, and $\operatorname{adj}(A)=\mathbf{0}$. In both cases, the entry (v,u) of $\operatorname{adj}(A)$ equals its diagonal entry d_u; hence $\operatorname{adj}(A)=\mathbf{1}d^\top$, and $\operatorname{adj}(A)A=\mathbf{0}$ gives $d^\top A=\mathbf{0}^\top$.

(b) We expand the determinant by linearity in each column, choosing $A_{\cdot v}$ or $c_v a$. The terms choosing $c_v a$ in two columns vanish. By (2.1) and part (a), we have

$$\det(A+ac^\top)=\det A+\sum_v c_v\sum_u a_u\operatorname{adj}(A)_{vu}=\sum_v c_v\sum_u a_u d_u=(c^\top\mathbf{1})(a^\top d).$$

∎

Proof of Lemma 2.6. We fix $v\in B^c$ and let $q_0\ge1$ be the first index with $f^{(q_0)}(v)\in B$. We first prove the claim that the iterates $f^{(q)}(v)$, $0\le q\le q_0$, are pairwise distinct: a repetition $f^{(q)}(v)=f^{(q')}(v)$ with $q<q'\le q_0$ forces both indices below q_0, because an element of B^c cannot equal one of B, and then makes the iterates of $f^{(q)}(v)$ periodic inside B^c, so that they never reach B. It follows that $v\in D_u$ exactly for $u=f^{(q)}(v)$, $0\le q\le q_0$, so v lies in the block of exactly one root, namely $f^{(q_0)}(v)$. Moreover, $v\in D_u\setminus\{u\}$ if and only if $f^{(q)}(v)=u$ for some $q\ge1$, that is, $v\in D_w$ for the child $w:=f^{(q-1)}(v)$ of u, which is unique by distinctness. In addition, $u\notin D_w$ for a child w of u, since otherwise the iterates from u would return to u. This completes the proof. ∎

Proof of Lemma 3.3. We first note that $(L\mathbf{1})_u=1-\pi_u^{-1}(P^\top\pi)_u=0$. Since $L[U]=\Pi[U]^{-1}\big((I-P)[U]\big)^\top\Pi[U]$, the two matrices have the same determinant, and

$$\big((I-P)[U]\big)^{-1}=\Pi[U]^{-1}\big(L[U]^{-1}\big)^\top\Pi[U],$$

whose (u,v) entry is $(\pi_v/\pi_u)(L[U]^{-1})_{vu}$.

We now fix $u\in U$ and $w\in U^c$. By irreducibility, there is a path of pairwise distinct states from u to w with positive transition probabilities, and it leaves U within at most $n-1$ steps. Hence the walk started at u leaves U within $n-1$ steps with positive probability, so every row of $P[U]^{n-1}$ sums to less than one, and the substochastic matrix $P[U]$ has spectral radius less than 1. Therefore, the Neumann series converges to $\big((I-P)[U]\big)^{-1}$, which proves (3.7). Moreover, the eigenvalues η of $P[U]$, lying in the open unit disk and closed under conjugation, give $\det\big((I-P)[U]\big)=\prod_\eta(1-\eta)>0$, the product over the eigenvalues with multiplicity.

The first identity in (3.8) is Cramer's rule. For the second, the column v of $L[U]^{-1}$, extended by zero off U and divided by its value at v, is 1 at v, vanishes off U, and is annihilated by the rows $U\setminus\{v\}$ of L, because $L[U]\,(L[U]^{-1})_{\cdot v}=e_v$; it is therefore $\gamma^{U\setminus\{v\}}(e_v)$ by uniqueness. This completes the proof. ∎

Proof of Corollary 4.9. We extend p to V by positive values on B and normalize its sum to one; clearly, both sides of (4.4) are unchanged. We set $L:=I-\mathbf{1}p^\top$.

Then L has zero row sums, $\det L[U] = 1 - p(U) > 0$ for every proper U by (2.1), and $L[U]\mathbf{1} = (1 - p(U))\mathbf{1}$; hence, for $v \in U^c$,

$$\Gamma^U_{\cdot v} = p_v L[U]^{-1}\mathbf{1} = \frac{p_v}{p(U^c)}\mathbf{1}.$$

Therefore, the factor at ℓ is $p_{\tau_\ell}/p(B^c_{\ell-1})$ when $f(\tau_{\ell+1}) \neq \tau_\ell$, and 1 otherwise. On the right, with $q_u := p(D_u)/p(B^c)$ for $u \in B$, we have $\Gamma^B M_f = \mathbf{1}q^\top$ and, by (2.1),

$$\det\big((I - \mathbf{1}q^\top)[B \setminus \{r\}]\big) = 1 - \sum_{u \neq r} q_u = q_r,$$

so Theorem 4.5 gives (4.4). For uniform p and $B = \{r\}$, multiplying by $k! = \prod_{\ell=1}^{k-1}(k - \ell + 1)$ turns each summand into the summand of (4.5). ∎

Appendix C. The identity for real aggregations.

The order-mass identity depends on the forest through its root indicator. We now extend it to any real aggregation with unit row sums. This extension sums over all orderings of the nonroot vertices, so it requires nonsingularity on every proper base containing the roots. We then remove the root set by doubling the vertex set, and quantify the defect when one aggregation row does not sum to one.

Assumption C.1 (All bases). *$L\mathbf{1} = \mathbf{0}$, and $\det L[U] \neq 0$ for every U with $B \subseteq U \subsetneq V$.*

Corollary C.2 (Real aggregation). *Let $B \subsetneq V$ be nonempty, $k := |B^c|$, let L satisfy Assumption C.1, and let $M \in \mathbb{R}^{B^c \times B}$ satisfy $M\mathbf{1} = \mathbf{1}$. Then, for every $r \in B$,*

$$\sum_{\tau} M_{\tau_1 r} \prod_{\ell=1}^{k-1} \sum_{u \in B} M_{\tau_{\ell+1} u}\, \Gamma^{B_{\ell-1}}_{u\tau_\ell} = \det\big((I - \Gamma^B M)[B \setminus \{r\}]\big), \tag{C.1}$$

the outer sum over all $k!$ orderings τ of B^c, with $B_\ell := B \cup \{\tau_1, \dots, \tau_\ell\}$.

Proof. We first consider the case in which every row of M is a standard basis vector, $M_{v\cdot} = e^\top_{f(v)}$ for a map $f : B^c \to B$. Then M is the root indicator of the B-parent map f, whose growth orders are all orderings of B^c and whose bases are all sets $B \subseteq U \subsetneq V$; the left side of (C.1) is $g_r(B, f)$, since $M_{\tau_1 r} = [f(\tau_1) = r]$ and the inner sum is $\tilde{\Gamma}^{B_{\ell-1}}_{f(\tau_{\ell+1})\tau_\ell}$, so (C.1) is Theorem 4.5.

For general M, we induct on the number of rows that are not standard basis vectors. The key step is that both sides of (C.1) are affine functions of any one row $m := M_{v\cdot}$ when the other rows are fixed. On the left, each term contains the row v exactly once, as $M_{\tau_1 r}$ if $\tau_1 = v$, and in the inner sum at the ℓ with $\tau_{\ell+1} = v$ otherwise. On the right, we have $I - \Gamma^B M = N_0 - \Gamma^B_{\cdot v} m^\top$ with N_0 independent of m, and a determinant whose columns are affine in m along one fixed vector is affine in m, since the terms with two such columns vanish. Now suppose that the row v is not a standard basis vector. Then the difference of the two sides, as a function of m on the hyperplane $\{\mathbf{1}^\top m = 1\}$, is affine and vanishes at the e_u, $u \in B$, by the induction hypothesis. Since $m = \sum_u m_u e_u$ with $\sum_u m_u = 1$, the difference at m is $\sum_u m_u$ times its values at e_u, which is zero. This completes the proof. ∎

The aggregation identity also yields a formula without a prescribed root set. Add the vertices in each possible order. At each step, pair the aggregation row of the

next vertex with the full harmonic extension of the current vertex into those already added. Summing these products gives a principal cofactor of $I - M$. The matrix L determines the individual terms but disappears from their sum.

PROPOSITION C.3 (The identity without a root set). *Let $L \in \mathbb{R}^{V\times V}$ satisfy $L\mathbf{1} = \mathbf{0}$ and $\det L[U] \neq 0$ for every U with $\emptyset \neq U \subsetneq V$, and let $M \in \mathbb{R}^{V\times V}$ have unit row sums. Then, for every $u \in V$,*

$$\sum_{\tau} M_{\tau_1 u} \prod_{\ell=1}^{n-1} \sum_{v\in V} M_{\tau_{\ell+1} v}\, \tilde{\Gamma}^{U_{\ell-1}}_{v\tau_\ell} = \det\big((I-M)[V\setminus\{u\}]\big), \tag{C.2}$$

the sum over all $n!$ orderings τ of V, with $U_\ell := \{\tau_1,\dots,\tau_\ell\}$; the factor $\ell = 1$ is $M_{\tau_2\tau_1}$, since $\tilde{\Gamma}^{\emptyset} = I$.

Proof. The key step is to double the vertex set. Let $B := \{1',\dots,n'\}$ be a copy of V and $V' := B \sqcup V$. In the block order (B, V), we let $L' := \left(\begin{smallmatrix} I_n & -I_n \\ \mathbf{0} & L \end{smallmatrix}\right)$, and read M as an aggregation of V into B. Then Assumption C.1 holds for (L', B): $L'\mathbf{1} = \mathbf{0}$; block triangularity gives $\det L'[B\cup U] = \det L[U] \neq 0$ for nonempty proper $U \subset V$; and $L'[B] = I_n$ handles $U = \emptyset$.

On the right side of (C.1), the matrix Γ^B of B for L' is $-L'[B]^{-1}L'[B,V] = I$, so the aggregated matrix is $I - M$, whose cofactor at u' is the right side of (C.2).

On the left side of (C.1), for a proper $U \subset V$ and $w \in U^c$, the harmonic extension x of e_w into $B\cup U$ for L' is defined by $x|_{V\setminus U} = e_w|_{V\setminus U}$, $x_{v'} = x_v$ for $v \in V$, and $L[U,V]\,x|_V = \mathbf{0}$; uniqueness gives $x|_V = \gamma^U(e_w)$ by Lemma 2.3, which for $U = \emptyset$ reads $x|_V = e_w$, and $x|_B$ is its copy. For $U = U_{\ell-1}$ and $w = \tau_\ell$ each inner factor of (C.1) becomes $\sum_{v\in V} M_{\tau_{\ell+1}v}\tilde{\Gamma}^{U_{\ell-1}}_{v\tau_\ell}$, and the initial factor is $M_{\tau_1 u'} = M_{\tau_1 u}$. Therefore, Corollary C.2 for (L', B, M) at u' is exactly (C.2). ∎

The affinity used in the proof of Corollary C.2 also gives the exact effect of a row with another sum.

PROPOSITION C.4 (The defect of a non-unit row). *Let L satisfy Assumption C.1, fix $v \in B^c$ and $r \in B$, and let $M \in \mathbb{R}^{B^c\times B}$ have unit row sums in every row other than the row v. As functions of $m := M_{v\cdot}$, the two sides of (C.1) satisfy*

$$\mathrm{LHS}(m) - \mathrm{RHS}(m) = \big(\mathbf{1}^\top m - 1\big)\cdot \det\big((I - \Gamma^B M^0)[B\setminus\{r\}]\big),$$

where M^0 is M with its row v replaced by $\mathbf{0}$.

Proof. The difference $h := \mathrm{LHS} - \mathrm{RHS}$ is affine in m, say $h(m) = a^\top m + a_0$, and vanishes on $\{\mathbf{1}^\top m = 1\}$ by Corollary C.2. For m_0 on that hyperplane and z with $\mathbf{1}^\top z = 0$, we have $a^\top z = h(m_0 + z) - h(m_0) = 0$; hence $a = c\mathbf{1}$ for a scalar c, and $h(m_0) = 0$ gives $a_0 = -c$. At $m = \mathbf{0}$, every term of the left side contains a factor linear in m, so $\mathrm{LHS}(\mathbf{0}) = 0$ and $c = -h(\mathbf{0}) = \mathrm{RHS}(\mathbf{0})$, which is the displayed cofactor. This completes the proof. ∎

Hence, for fixed other rows and a fixed root r, (C.1) holds for a row with $\mathbf{1}^\top m \neq 1$ if and only if the cofactor $\det\big((I-\Gamma^B M^0)[B\setminus\{r\}]\big) = 0$. Unit row sums are therefore sufficient for every configuration and every root, and they cannot be dropped in general, but they are not necessary for a fixed configuration and root. For example, let $V = \{1,2,3,4\}$, $B = \{1,2\}$, $L = I - \mathbf{1}\mathbf{1}^\top/4$, $M_{3\cdot} = (a,b)$ and $M_{4\cdot} = (-1,2)$. Then every entry of Γ^B equals $\frac{1}{2}$, the cofactor vanishes at $r = 1$, and both sides of (C.1) equal $-\frac{b}{2}$ at $r = 1$ for all a, b; at $r = 2$, the cofactor is $\frac{3}{2}$, and the difference of the two sides is $\frac{3}{2}(a + b - 1)$.

The zero row sums of L cannot be dropped either. For example, for $|B| = 2$, $|B^c| = 1$ and $f(w) = 1$, the cofactor at the root 2 is $1 - \Gamma^B_{1w}$, which vanishes by (2.2), as the left side of (4.2) at $r = 2$ does. On the other hand, without zero row sums in the rows B, the entry $\Gamma^B_{1w} = -(L[B]^{-1}L[B, \{w\}])_1$ is arbitrary.

Appendix D. Randomizing the seed does not remove the tilt.

By Theorem 6.4, the stopped covering walk samples the weight-proportional law tilted by the completion weight. We now show that mixing seeds cannot remove the tilt in general.

PROPOSITION D.1 (No seed mixture is unbiased in general). *Let P be irreducible and stochastic with stationary distribution π, let λ be as in (3.4), let $B \subsetneq V$ be nonempty, and for $r \in B$ let $\mathbb{P}_r$ be the law of the first-entrance forest of the stopped covering walk from r. For every $c \in \mathbb{R}^B$ and every $F_0 \in \mathcal{F}_B$ with parent map f,*

$$\sum_{r \in B} c_r \, \mathbb{P}_r(F_B = F_0) = \frac{w(F_0)}{(\mathbf{1}^\top \lambda) \prod_{v \in V} \pi_v} \sum_{r \in B} c_r \, \varphi_r(F_0). \tag{D.1}$$

For the chain of Theorem A.1 with $B = \{1, 2\}$, no $c \neq \mathbf{0}$ makes the left side proportional to the forest weight on $\mathcal{F}_B$.

Proof. Formula (D.1) follows immediately from (6.2). In the example, every forest has positive weight, so proportionality to the forest weight forces $\sum_r c_r \varphi_r(F_0)$ to take the same value for every $F_0 \in \mathcal{F}_B$. According to Table 1, (φ_1, φ_2) equals $(\frac{1}{5}, \frac{1}{20})$, $(\frac{1}{20}, \frac{1}{5})$ and $(\frac{3}{20}, \frac{3}{20})$ at f_1, f_2 and f_3, respectively. Equating the values at f_1 and f_2 gives $c_1 = c_2 =: c$. The values at f_1 and f_3 are then $(\frac{1}{5} + \frac{1}{20})c = \frac{c}{4}$ and $(\frac{3}{20} + \frac{3}{20})c = \frac{3c}{10}$, forcing $c = 0$. This completes the proof. ∎

To explain the gist of the tilt, let $B = \{r, u\}$. A completion of F_0 adds one edge into u from the tree of r, so $\varphi_r(F_0) = \sum_{x \in D_r \cup \{r\}} \pi_x P_{xu}$ is the stationary flow from the tree of r into u. Hence the walk from r favors the forests whose tree of r sends much stationary flow into u, and randomizing the seed only averages two such preferences.

Appendix E. Lean formalization.

We formalize the order-mass identity, the Aldous–Broder theorem, and the stopped-forest law in Lean 4 [6] with mathlib [26] (toolchain v4.30.0, mathlib v4.30.0), in the directory `EAB/Paper` of the repository https://github.com/YuZh98/aldous-broder-lean. The development imports mathlib only, follows the proofs of this article, and contains no `sorry`. The four endpoint theorems `order_mass_identity`, `Chain.aldous_broder`, `Chain.aldous_broder_normalized` and `Chain.stopped_forest_law`, together with their principal proof obligations, depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`, which `lake env lean AxiomCheck.lean` prints after `lake build`. We list the correspondence in Table 4; all declarations are in the namespace `EAB.Paper`.

The Lean statements differ from the text in a few modeling choices, which we now list. The law of the chain started at r is mathlib's Ionescu-Tulcea trajectory measure (`Chain.chainLaw`), checked against the cylinder formula (`Chain.chainLaw_cylinder`), and probabilities are extended nonnegative reals. The forest event requires the walk to visit every state, which has probability one (`Chain.ae_hits_all`), and its first-entrance parents to equal f; this is the event $\{F_B = F_0\}$ up to a null set. The stationary distribution enters as a hypothesis that π is positive, normalized and invariant, and `Chain.IsStationary.cofactorVec_eq` proves that such a π is the normalized cofactor vector, hence unique. The standing

TABLE 4
Results and their Lean declarations.

result	Lean declaration
Definition 2.2, (2.2)	`Foundation.harmonic`, `Foundation.gamma`, `Foundation.harmonic_unique`, `Foundation.harmonic_one`
(2.3)	Γ^U_{uv} is `Foundation.gamma L U v u` for $u \in U$
Lemma 2.1	`Foundation.adjugate_apply_eq_principalCofactor`, `Foundation.principalCofactor_left_kernel`, `Foundation.det_add_rank_one_zero_row_sum`
Definition 2.5, Lemma 2.6	`IsPaperParentMap`, `growthOrders`, `rootIndicator`, `rootBlock_pairwiseDisjoint`, `rootBlocks_cover`, `childDescendants_cover`
(3.3), Definition 4.3	`orderWeight`, `orderMass`
Lemma 5.1	`Foundation.one_vertex_transfer`, `Foundation.one_vertex_transfer_singleton`
Theorem 4.5	`order_mass_identity`, `single_root_normalization`
Lemma 3.3	`Chain.det_principal_reversedMatrix_pos`, `Chain.green_hasSum`, `Chain.green_reversal`, `Chain.inv_diag_eq_det_ratio`, `Chain.inv_column_eq_gamma`
Lemma 3.4	`Chain.isStationary_cofactor`, `Chain.IsStationary.cofactorVec_eq`
Proposition 3.2	`Chain.chainLaw_orderEvent_eq`
Theorem 6.1	`Chain.aldous_broder`, `Chain.aldous_broder_normalized`
Theorem 6.4	`Chain.stopped_forest_law`

assumption $n \geq 2$ appears as `2 ≤ Fintype.card V`. Theorem 4.5 is stated over an arbitrary field, and `Chain.stopped_forest_law` holds for every parent assignment, including those of weight zero; it states (6.3) together with the law obtained by substituting (6.3) into (6.2). Lemma 2.6 is formalized for the children of roots, which is the case the proof uses, and `Foundation.one_vertex_transfer` allows a fold along any weights that sum to one, of which Lemma 5.1 is the fold into a single root.

Proposition 3.2 is formalized for growth orders in its general form (3.5); its statement for the other orderings and the simplification (3.6) are not stated separately. Finally, we have not formalized Theorem 6.2, Corollary 6.6, Corollary 4.9, the results of sections C and D, and the numerical experiments.

Appendix F. Numerical illustration.

We use small finite examples to examine three consequences of the stopped-forest law: its distance from the weight-proportional law, the target mass lost through incomplete support, and the variability of importance-corrected estimates. We generate 30 chains in each of three families on $V = \{1, \dots, 7\}$. Complete enumeration of the forest space gives both laws without sampling. The experiments illustrate the formulas and check their implementation; they do not estimate typical bias on larger graphs.

F.1. Setting. Table 5 specifies the dense, graph, and drift families. For every chain, we use the nested root sets $B = \{1\}, \{1, 2\}, \dots, \{1, \dots, 6\}$ with seed 1. Thus the root-count comparisons reuse the same chain.

In the drift family, changing the holding weights leaves the jump chain unchanged and multiplies every off-diagonal stationary flow $\pi_u P_{uv}$ by a common constant. The 30 drift chains therefore share the same forest laws and give one example for the distributional comparisons. We display that example once. Their holding weights do affect the time needed to cover the nonroot vertices.

For each chain and root set, we enumerate the parent maps in $\mathcal{F}_B$. We compute

TABLE 5
The three families of chains on $V = \{1, \dots, 7\}$, with 30 chains each.

family	transition matrix	reversible
dense	$P_{uv} = W_{uv} / \sum_x W_{ux}$, with W_{uv} independent and uniform on $\{1, \dots, 5\}$	in general no
graph	simple random walk on a random graph, in which each of the 21 possible edges is present independently with probability 0.45, before conditioning on connectivity	yes
drift	$P_{uv} \propto W_{uv}$, with $W_{u,u+1} = 3$ and $W_{u,u-1} = 1$ (indices modulo 7), and W_{uu} uniform on $\{0, 1, 2\}$	no

the order masses from the principal cofactors, then obtain the stopped-walk law q_A with seed 1 from (6.2). The comparison law is $q_W \propto w$. We measure their difference by total variation distance, $\frac{1}{2} \sum_f |q_A(f) - q_W(f)|$. The stationary distributions and normalizing constants are computed in exact rational arithmetic; the remaining linear algebra uses double precision.

Support loss requires distinguishing structural zeros from numerical roundoff. By Theorem 6.7, $\varphi_1(F_0) = 0$ exactly when some root cannot reach 1 through positive entries of Q. The script zero_check.py compares this combinatorial test with a 10^{-14} threshold on the factor $g(B, f)^\top L[B]^{-1} e_1$, which differs from the completion weight by a positive constant for each fixed chain and root set. The tests agree in all 546,120 cases. Computed values of this factor at structural zeros are at most 2.2×10^{-16} in magnitude, and the smallest positive value is 2.7×10^{-2}.

The script make_experiments.py generates the chains and enumerated laws with random seed 20260918. The figure script plot_bias.py reads these stored outputs to produce Figure 5, without recomputing the laws. All scripts named in this appendix and their stored outputs are in the repository https://github.com/YuZh98/aldous-broder-numerics.

F.2. Results. For one root, the two laws agree, as Theorem 6.1 requires. With two roots, the median total variation distance is about 0.26 for both the dense and graph families. With three roots, the medians are about 0.27 and 0.30, respectively. The largest distance is 0.75, attained by one graph chain with four roots. Beyond three roots, the median distances decrease to about 0.11 at $|B| = 6$. These examples show that a stopped walk can differ substantially from the target forest law even when the underlying chain is reversible, as in the graph family.

The drift example has smaller distances for the chosen roots and seed, increasing from 0.001 at $|B| = 2$ to 0.083 at $|B| = 6$. This behavior depends on the contiguous roots and seed 1; the next experiment examines that dependence. All reported magnitudes concern these chain families and root choices.

F.3. Sensitivity. We vary the placement of the roots, the strength of the drift and the connectivity of the graph; the script robustness.py reproduces the values below.

Placement. For the same 30 dense and 30 graph chains, we replace the nested roots and the seed 1 by a uniformly random root set of each size and a uniformly random seed in it. The median distances for $|B| = 2, \dots, 6$ are 0.24, 0.29, 0.21, 0.20 and 0.13 for the dense family and 0.23, 0.28, 0.30, 0.27 and 0.16 for the graph family, close to those of Figure 5, since these families do not distinguish the labels of the

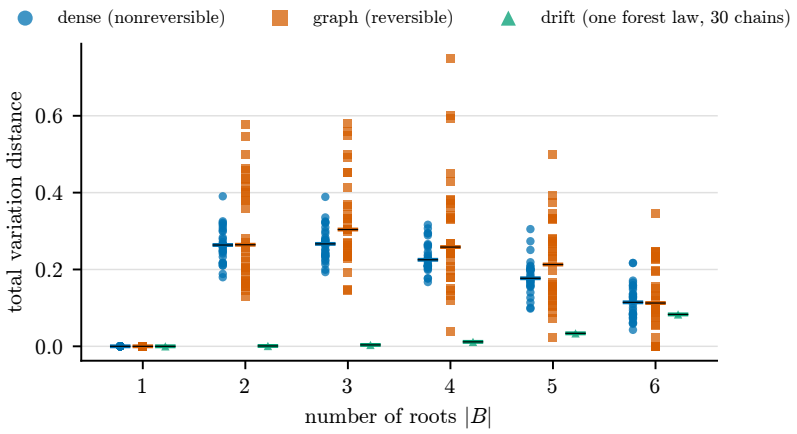


FIG. 5. *Total variation distance between the stopped-walk forest law (6.2) with seed* 1 *and the weight-proportional law, for the three families of Table 5 and root sets* $B = \{1\}, \{1,2\}, \ldots, \{1,\ldots,6\}$. *Each point is one chain, and the horizontal bars mark the medians; the drift family is shown once, since its* 30 *chains share one forest law. For* $|B| = 1$, *the distance is zero by Theorem 6.1.*

states.

Drift. The cycle does distinguish them. Table 6 gives the range of the distance over the seeds in the root set, for contiguous roots $\{1,\ldots,b\}$ and for roots spread evenly around the cycle, and for three drift strengths. The small distances of Figure 5 occur only for the contiguous roots and one seed; with another seed, or with spread roots, the distance is of the same order as for the other families.

Connectivity. For 30 random connected graphs with edge probability 0.30, 0.45 and 0.70, the median distance for two roots is 0.40, 0.23 and 0.23, and the fraction of chains with lost mass is 0.57, 0.23 and 0.13; sparser graphs give larger distances and more lost mass.

TABLE 6

Range of the total variation distance over the seeds in the root set, for the drift cycle with forward weight a *and backward weight* 1.

a	roots	$\lvert B\rvert = 2$	3	4	5	6
1.5	contiguous	0.03–0.23	0.07–0.36	0.13–0.44	0.16–0.45	0.01–0.37
	spread	0.46	0.56	0.48–0.51	0.26–0.39	0.01–0.37
3	contiguous	0.00–0.22	0.00–0.29	0.01–0.31	0.03–0.30	0.08–0.25
	spread	0.30–0.31	0.43–0.44	0.40–0.47	0.24–0.40	0.08–0.25
6	contiguous	0.00–0.14	0.00–0.16	0.00–0.17	0.01–0.16	0.02–0.14
	spread	0.16	0.27–0.28	0.26–0.34	0.14–0.26	0.02–0.14

F.4. Lost mass, effective sample size and an estimator. Total variation distance alone does not identify whether correction is possible. We therefore report two diagnostics in Table 7: the unreachable target mass $q_W(S_1^c)$ and the limiting normalized effective sample size of the inverse-completion weights. The former measures the part of the target law that no reweighting can recover. The latter measures weight dispersion on the reachable forests. For m independent draws $F^{(i)} \sim q_A$ with weights

$a_i := 1/\varphi_1(F^{(i)})$, the normalized effective sample size $(\sum_i a_i)^2/(m\sum_i a_i^2)$ converges almost surely to

$$\mathrm{ESS}/m := \frac{(\mathbb{E}_{q_A} a)^2}{\mathbb{E}_{q_A} a^2} = \Big(\sum_{F_0 \in S_1} \frac{\bar{q}_W(F_0)^2}{q_A(F_0)} \Big)^{-1}, \qquad \bar{q}_W := q_W(\cdot \mid S_1),$$

which is the value reported in the table. Its interpretation concerns weight dispersion: the precision of a particular estimate also depends on the function h and its association with the weights.

Lost mass is zero for all dense and drift chains. In the graph family, it is positive for about half the chains with two or three roots and reaches 0.75. For the graph chain with four roots that attains the largest total variation distance in Figure 5, the distance is entirely lost mass. Reweighting cannot reduce that distance. On the reachable forests, the median limiting normalized ESS is between 0.59 and 0.95 for the dense and graph families, with a minimum of 0.31.

TABLE 7

Lost target mass $q_W(S_1^c)$ for the graph family and limiting normalized effective sample size ESS/m *for the three families, over the* 30 *chains of each family. The lost mass is zero for every dense and drift chain.*

	graph: lost mass			ESS/m				
$\lvert B\rvert$	fraction > 0	median	max	dense med.	dense min	graph med.	graph min	drift
2	0.53	0.15	0.58	0.59	0.31	0.83	0.69	1.00
3	0.57	0.10	0.58	0.68	0.43	0.73	0.52	1.00
4	0.37	0	0.75	0.79	0.59	0.76	0.34	1.00
5	0.20	0	0.50	0.86	0.72	0.87	0.47	0.99
6	0.03	0	0.20	0.94	0.83	0.95	0.62	0.97

To illustrate the effect on an estimate, consider the expected size of the seed's tree, $h(F_0) := 1 + |D_1|$, with $B = \{1, 2\}$. The first dense chain has complete support. Its target expectation is 3.237, while the uncorrected stopped-walk expectation is 4.359. The self-normalized correction converges to 3.237. Across 4,000 replications with $m = 200$ independent draws each, the corrected estimate has mean 3.241 and standard deviation 0.175; the limiting normalized ESS is 0.68.

For the graph chain with the largest lost mass, 0.577, the target expectation is 1.923 and the uncorrected expectation is 3.733. The correction converges to the conditional target 3.182. Thus correct weighting on the sampled forests does not resolve the loss of the other forests. These calculations are generated by estimator_demo.py, which reuses the chains from make_experiments.py and uses random seed 1 for the replications.

F.5. Checks. We check the formulas first by enumeration. For every chain, root set, and seed, the computed stopped-walk probabilities sum to one and agree with the tree-law marginal obtained by enumerating the arborescences. The largest discrepancy is 4.8×10^{-15}. These checks verify the algebraic evaluations against a separate enumeration of completed trees.

We also run Algorithm 3.1 200,000 times on the first dense chain with $|B| = 2$ and seed 1. This simulation uses only the forward transition matrix. Of the 4,802 expected counts, 755 are below 5 and 79 below 1; the smallest is 0.26. We therefore calibrate the goodness-of-fit statistics with 2,000 multinomial samples of the same size

from q_A, rather than a chi-squared approximation. The reported upper-tail p-values have Monte Carlo standard errors at most 0.011.

The Pearson statistic is 4,818.1, with simulated p-value 0.41. The maximum standardized deviation across forests is 3.52, with upper-tail p-value 0.995. This maximum lies near the lower 0.5% tail of the simulated reference distribution and is unusually small under q_A. The upper-tail test does not reject, but the small maximum should not be read as evidence of better agreement. The enumeration and the forward-walk simulation provide different checks; the simulation statistics do not replace the deterministic checks.

REFERENCES

[1] D. J. Aldous, *The random walk construction of uniform spanning trees and uniform labelled trees*, SIAM Journal on Discrete Mathematics, 3 (1990), pp. 450–465, https://doi.org/10.1137/0403039.

[2] V. Anantharam and P. Tsoucas, *A proof of the Markov chain tree theorem*, Statistics & Probability Letters, 8 (1989), pp. 189–192, https://doi.org/10.1016/0167-7152(89)90016-3.

[3] L. Avena and A. Gaudillière, *Two applications of random spanning forests*, Journal of Theoretical Probability, 31 (2018), pp. 1975–2004, https://doi.org/10.1007/s10959-017-0771-3.

[4] A. Z. Broder, *Generating random spanning trees*, in Proceedings of the 30th Annual Symposium on Foundations of Computer Science (FOCS), IEEE, 1989, pp. 442–447, https://doi.org/10.1109/SFCS.1989.63516.

[5] S. Chaiken, *A combinatorial proof of the all minors matrix tree theorem*, SIAM Journal on Algebraic and Discrete Methods, 3 (1982), pp. 319–329, https://doi.org/10.1137/0603033.

[6] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in Automated Deduction – CADE 28, vol. 12699 of Lecture Notes in Computer Science, Springer, 2021, pp. 625–635, https://doi.org/10.1007/978-3-030-79876-5_37.

[7] L. L. Duan and A. Roy, *Spectral clustering, Bayesian spanning forest, and forest process*, Journal of the American Statistical Association, 119 (2024), pp. 2140–2153, https://doi.org/10.1080/01621459.2023.2250098.

[8] S. Fomin, *Loop-erased walks and total positivity*, Transactions of the American Mathematical Society, 353 (2001), pp. 3563–3583, https://doi.org/10.1090/S0002-9947-01-02824-0.

[9] L. Fredes and J.-F. Marckert, *A combinatorial proof of Aldous–Broder theorem for general Markov chains*, Random Structures & Algorithms, 62 (2023), pp. 430–449, https://doi.org/10.1002/rsa.21101.

[10] R. A. Horn and C. R. Johnson, *Matrix Analysis*, Cambridge University Press, Cambridge, 2nd ed., 2013, https://doi.org/10.1017/CBO9781139020411.

[11] Y. Hu, R. Lyons, and P. Tang, *A reverse Aldous–Broder algorithm*, Annales de l'Institut Henri Poincaré, Probabilités et Statistiques, 57 (2021), pp. 890–900, https://doi.org/10.1214/20-AIHP1101.

[12] A. Kruckman, A. Greenwald, and J. Wicks, *An elementary proof of the Markov chain tree theorem*, Tech. Report CS-10-04, Brown University, Department of Computer Science, 2010.

[13] F. T. Leighton and R. L. Rivest, *Estimating a probability using finite memory*, IEEE Transactions on Information Theory, 32 (1986), pp. 733–742, https://doi.org/10.1109/TIT.1986.1057250.

[14] D. A. Levin and Y. Peres, *Markov Chains and Mixing Times*, American Mathematical Society, Providence, RI, 2nd ed., 2017, https://doi.org/10.1090/mbk/107.

[15] R. D. Luce, *Individual Choice Behavior: A Theoretical Analysis*, Wiley, New York, 1959.

[16] Z. T. Luo, H. Sang, and B. Mallick, *A Bayesian contiguous partitioning method for learning clustered latent variables*, Journal of Machine Learning Research, 22 (2021), pp. 1–52, https://jmlr.org/papers/v22/20-136.html.

[17] R. Lyons and Y. Peres, *Probability on Trees and Networks*, vol. 42 of Cambridge Series in Statistical and Probabilistic Mathematics, Cambridge University Press, Cambridge, 2016, https://doi.org/10.1017/9781316672815.

[18] P. Marchal, *Loop-erased random walks, spanning trees and Hamiltonian cycles*, Electronic Communications in Probability, 5 (2000), pp. 39–50, https://doi.org/10.1214/ECP.v5-1016.

[19] I. Nunes, G. Iacobelli, and D. R. Figueiredo, *A transient equivalence between Aldous–*

Broder and Wilson's algorithms and a two-stage framework for generating uniform spanning trees, Discrete Mathematics, 348 (2025), https://doi.org/10.1016/j.disc.2024.114285. Art. 114285.

[20] Y. Y. Pilavci, P.-O. Amblard, S. Barthelmé, and N. Tremblay, *Graph Tikhonov regularization and interpolation via random spanning forests*, IEEE Transactions on Signal and Information Processing over Networks, 7 (2021), pp. 359–374, https://doi.org/10.1109/TSIPN.2021.3084879.

[21] J. Pitman and W. Tang, *Tree formulas, mean first passage times and Kemeny's constant of a Markov chain*, Bernoulli, 24 (2018), pp. 1942–1972, https://doi.org/10.3150/16-BEJ916.

[22] R. L. Plackett, *The analysis of permutations*, Journal of the Royal Statistical Society. Series C (Applied Statistics), 24 (1975), pp. 193–202, https://doi.org/10.2307/2346567.

[23] P. Pokarowski, *Directed forests with application to algorithms related to Markov chains*, Applicationes Mathematicae, 26 (1999), pp. 395–414, https://doi.org/10.4064/am-26-4-395-414.

[24] J. G. Propp and D. B. Wilson, *How to get a perfectly random sample from a generic Markov chain and generate a random spanning tree of a directed graph*, Journal of Algorithms, 27 (1998), pp. 170–217, https://doi.org/10.1006/jagm.1997.0917.

[25] E. Tam, D. B. Dunson, and L. L. Duan, *Exact sampling of spanning trees via fast-forwarded random walks*, Biometrika, 112 (2025), https://doi.org/10.1093/biomet/asaf031. Art. asaf031.

[26] The mathlib Community, *The Lean mathematical library*, in Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs, ACM, 2020, pp. 367–381, https://doi.org/10.1145/3372885.3373824.

[27] W. T. Tutte, *The dissection of equilateral triangles into equilateral triangles*, Proceedings of the Cambridge Philosophical Society, 44 (1948), pp. 463–482, https://doi.org/10.1017/S030500410002449X.

[28] A. D. Ventsel' and M. I. Freidlin, *On small random perturbations of dynamical systems*, Russian Mathematical Surveys, 25 (1970), pp. 1–55, https://doi.org/10.1070/RM1970v025n01ABEH001254. Translated from Uspekhi Mat. Nauk, 25 (1970), pp. 3–55.

[29] D. B. Wilson, *Generating random spanning trees more quickly than the cover time*, in Proceedings of the 28th Annual ACM Symposium on Theory of Computing (STOC), ACM, 1996, pp. 296–303, https://doi.org/10.1145/237814.237880.